

Information Revealed through the Regulatory Process: Interactions between the SEC and Companies ahead of Their IPO

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We analyze communications between the SEC and firms prior to IPOs using LDA analysis and KL divergence. The SEC's concerns closely map onto the regulator's stated mandate: companies increase prospectus disclosures on precise topics of SEC concern. Revenue recognition is the dominant topic of SEC concern, and it is not independently discovered by investors. Increased SEC concern about it is associated with greater secondary sales, lower post-IPO liquidity, lower post-IPO returns, and a higher probability of withdrawal. The regulator's role during the capital raising process results in increased transparency but contributes to delays in the going public process. (*JEL* G30, G34, G38)

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Introduction

Financial markets regulations are costly. However, the costs associated with insufficient regulations can be equally severe. The balancing act between the benefits of increased transparency, price informativeness and fairness versus the costs of regulatory requirements makes the role of the regulator a subject of continual debate. Although this tension is pervasive throughout markets,

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it is particularly strong when public information is limited and when there is greater separation of ownership and control. The objectives of this paper are twofold. First, we document the regulator's concerns at a point when its role is particularly salient, around the time a company prepares to raise public equity for the first time, and we examine the company's responses to these concerns. Second, we study the contribution of the regulatory process to the firm's information environment and to the firm's valuation.

We focus on communications between the Securities and Exchange Commission (SEC) and the company prior to a major capital raising event, the initial public offering (IPO). We examine the extent to which SEC concerns influence the going public process, the extent to which they are related to company value, and the extent to which they are independently uncovered by investors. First, we find that the SEC is not a passive player in the process, sending an average of 3.8 letters and with 41 questions in the initial letter to each firm in the pre-IPO period before the final prospectus is approved. About 70% of prospectus revisions are a result of SEC comment letters. Second, we find that the intensity of SEC review appears to influence both the amount of material a company has to disclose and the amount of time a company spends in registration. Third, we find that questions related to revenue recognition are most informative about company valuation and liquidity, and that investors do not fully realize these issues at the time of their IPO. Fourth, our findings raise questions regarding the trade-off between a company's "right to privacy" and increased transparency to potential investors.

Analyses of the communications between the IPO firm and the SEC reveal substantial variation, with the number of SEC letters to each company ranging from one to 14 and with substantial differences in topics of focus. We use a set of textual analysis techniques based on machine learning rather than human input to determine the number of distinct topics across these SEC letters. We identify the words and phrases that belong together within each topic, and to assess the economic meaning of each topic. In addition, we introduce a technique that enables us to match topics between two different types of documents, specifically between the initial disclosures of companies in the preliminary IPO prospectuses and the regulator comments that are conveyed through the SEC letters.¹ Our analysis indicates eight distinct topics of focus, which range from more boilerplate issues to questions on executive compensation and revenue recognition. Results further indicate that the SEC actively pursues its stated goals, with the main topics of SEC concerns mapping closely into its stated economic concerns.² The specific topics of SEC focus are also related to

¹ Machine learning techniques include LDA analysis and KL divergence, both of which are described in more detail in Section 1.3.

² The SEC's stated areas of focus include (a) a description of the company's business; (b) a description of the security to be offered for sale; (c) information about the management of the company; and (d) financial statements certified by independent accountants. See <https://www.sec.gov/fast-answers/answersregis33.htm>.

both company characteristics and to company disclosure choices in the first preliminary prospectus. For example, companies with higher likelihoods of accounting fraud receive more questions related to accounting topics, such as revenue recognition and capitalization.

Predicated on the Securities Acts of 1933 and 1934, a key objective of the regulator is to ensure sufficient information disclosure, thereby enabling market participants to have access to information and thus increase fairness and price informativeness (see, e.g., Coffee 1984). Legal scholars, such as Mahoney (1995), further emphasize the regulator's role in ensuring that the company does not engage in self-promotion, for example, by disclosing an upwardly biased set of information. Related to this, the SEC has a specific directive to protect financially unsophisticated investors (see, e.g., Hicks 1994), an objective that heightens the importance of regulators monitoring disclosures around the time of their IPO, when limited public information can cause uninformed investors to disproportionately receive allocations of "lemons" (see, e.g., Rock 1986).³

Motivated by these considerations, the second portion of the paper examines the ways in which the SEC's efforts influence both the transactions costs of going public and the company's information environment. Consistent with regulator influence, we find that SEC demands for additional disclosure result in companies disclosing significantly more information. Moreover, companies' increased disclosures closely relate to the topics on which the SEC had expressed concerns. This dynamic process of information disclosure arguably increases transparency in ways consistent with the SEC's objectives, but not without costs. Specifically, companies with more extensive SEC reviews experience delays in going public, with a significantly greater number of days between initial prospectus filing and ultimate IPO. Longer times in registration distract management from the running of the company, and the delays are risky due to the possibility of a deterioration in market conditions that causes the offering to be withdrawn (see, e.g., Dunbar and Foerster 2008).

Our third main finding is that information uncovered by the regulator is significantly related to estimates of firm value, liquidity, and costs associated with the going public process. Companies for which the SEC expresses more extensive concerns (measured by number of SEC letters) have significantly lower price revisions, fewer and smaller revisions of number of shares offered, and longer time spent at the registration. At the time of their IPO, these companies also have higher secondary sales; after their IPO, they have higher volatility and illiquidity; and around the time of the lockup expiration, they have greater insider sales. Results using latent Dirichlet allocation (LDA) analyses show the extent to which these relations are concentrated within certain topics of SEC concern. Conditional on the aggregate level of SEC concern, companies

³ In a speech, the SEC Chair Mary Jo White (2014) said "The retail investor must be a constant focus of the SEC – if we fail to serve and safeguard the retail investor, we have not fulfilled our mission."

with more questions related to capitalization, liquidity, and revenue recognition have significantly lower price updates. Moreover, across all analyses, revenue recognition concerns are the most salient. Companies about which the SEC expresses more concern on this single topic experience greater insider sales both at the time of their IPO and around the time of the lockup expiration, they have higher volatility and illiquidity, and they have lower post-IPO abnormal returns.

To examine the extent to which regulator's information production is unique, we take advantage of the fact that SEC letters are not made public until at least 20 days after their IPO (45 days during the earlier portion of our sample period). Thus, at the time of an IPO, the market has limited information about the specific issues raised by the regulator, specifically only what they are able to extract from analyzing changes in the prospectus content. If investors are not able to independently fully uncover these concerns, then such concerns will be followed by lower post-IPO returns when the content of the letters is revealed. Our findings indicate that while the market incorporates the information embedded in most topics of SEC concerns, concerns related to revenue recognition represent a notable exception. More extensive concerns on this topic are associated with significantly lower abnormal returns in the days surrounding release of the SEC letters, an effect that is concentrated among companies with greater investor downloads of these letters.

Our final set of results highlights the ways in which the regulator's information production is related to the set of companies that ultimately go public. While the regulator does not have the right to prevent a company from going public, the regulator's demands for information disclosure can cause a company to determine that an IPO is no longer the optimal course of action. To this end, we examine the relation between the extent of SEC concerns and the decision to withdraw an IPO from the market. Through a Freedom of Information Act (FOIA) request, we were able to obtain details on communications from the SEC and firms that ultimately withdraw their offerings. Consistent with expectations, we find that companies that receive more SEC concerns and that opt to disclose less in response to SEC questions are significantly more likely to ultimately withdraw their offerings. Further, companies where SEC concerns focus on revenue recognition exhibit a greater tendency to withdraw. While prior literature has highlighted the effects of market returns on IPO withdrawal, our results indicate that the SEC review process also plays a significant role.

Our paper contributes to several streams of literature. First, a broad body of literature, stemming from Rock (1986), Beatty and Ritter (1986), and Benveniste and Spindt (1989), considers the ways in which uncertainty and information asymmetry between the company and investors affect the valuation of IPOs. Following these theories, a stream of literature examines how the quality of information firms share with investors in the pre-IPO process affects key post-IPO outcomes (see, e.g., Hanley 1993; Hanley and Hoberg 2010, 2012;

Loughran and McDonald 2013). Our paper extends and complements these earlier efforts, by highlighting the influence of the regulator throughout this process. While companies expend considerable time and effort deciding how to portray their company to investors through the prospectus, the regulator frequently requires information beyond what the company initially chooses to disclose.

Our paper also relates to the ongoing debate in the literature over the role of the regulator. One stream of literature, starting with Stigler (1964), questions the necessity of the regulator, arguing that voluntary disclosures will lead to value maximizing outcomes.⁴ In contrast, Mahoney (1995) and others argue that in the absence of external monitoring, corporate issuers will tend to disclose an upwardly biased set of information. Our paper relates directly to this debate. Our findings indicate that the regulator plays a unique role. More regulatory intervention is associated with more information production by the firm, arguably decreasing the ability of companies to present an upwardly biased picture. Our conclusions are broadly consistent with Simon's (1989) analysis of companies around the time of the 1933 Securities Act and with LaPorta et al.'s (2006) international analysis; our study highlights the effects of these issues at a key point in the company's life cycle when information asymmetry is high and individual investors are particularly sensitive to lemons-type problems.

Our paper also fits into a small but growing number of papers that examine SEC comment letters to discern the value of the regulator. Cassell, Dreher and Myers (2013), Bozanic and Johnston (2017), Bozanic, Dietrich and Johnson (2017), Bozanic, Choudhary and Merkley (2017), Ryans (2018), and DeChow, Lawrence, and Ryans (2016) examine SEC comment letters to 10K filings of more mature firms; findings indicate that SEC letters predict problems within a firm, for example, restatements and negative performance, and they can also contribute to improved information environments. Also related to our work, Ertimur and Nondorf (2006) manually read and analyze letters of 130 IPOs. However, their small sample combined with a noisy categorization of topics limits their ability to identify the information content of the regulatory process. A contemporaneous paper by Li and Liu (2017) similarly looks at SEC letters to IPO firms, but they do not conduct any analysis of the topics on which the SEC focuses. The focus of their paper is more limited, and they do not examine the possible effects of regulator intervention, the uniqueness of the regulator's information production relative to investors, or the association between regulator demands for additional disclosure and offering withdrawals.

Finally, our paper introduces novel methodologies, which make several contributions to the growing literature using textual techniques. First, we use machine learning algorithms to not only identify individual topics and agnostically determine the number of topics but also discern the economic

⁴ Specifically, companies will choose to disclose up to the point at which the marginal benefits of additional disclosure equal the marginal costs.

meaning of each. Second, we use two distinct sets of LDA analyses combined with Kullback-Leibler (KL) divergence to analyze the marginal information contained in one document relative to another. Relative to Huang et al. (2018), who use a single LDA analysis to examine two document types (analyst reports and conference calls), our approach ensures that the shorter document type is not dominated by the longer type and more generally that topic identification in one set of documents is not contaminated by information contained within the other set. In our setting, this equates to ensuring that the SEC letter content is not dominated by prospectus content, and that we do not erroneously conclude that a topic was of great concern to the SEC when in fact it represented only a very small portion of the SEC's concern but a large portion of the prospectus. Third, our use of KL divergence to match topics, rather than cosine similarity, has advantages related to robustness of conclusions relative to the frequency of word use, as discussed further in the next section. Importantly, these methodologies could similarly apply to any case in which several documents have overlapping content but different formats, for example, company 10Ks and analyst reports, or company news releases and journalist reports.

1. SEC's Concerns and IPO Firms' Disclosures

1.1 IPO sample

We obtain a sample of firms that went public between May 2005 and December 2016 from Thomson Financial SDC Platinum. The beginning of our sample period is dictated by the fact that the SEC began disclosing all comment letters on May 12, 2005. Following the steps described in Lowry, Michaely, and Volkova (2017), we limit our sample to companies that went public on one of the U.S. stock exchanges, and we exclude real estate investment trusts (REITs), American depositary receipts (ADRs), unit offerings, and IPOs with an offer price below \$5.00. Additionally, we require all firms to have a CRSP record within 10 days from the offer date and to have a Compustat record 1 year prior to their IPO. From Thomson, we obtain characteristics of the offer, for example, the identity of the underwriter(s), the identity of the law firm(s), and whether it is venture backed. We calculate firm age using the year of founding from Jay Ritter's Web site.⁵ In the eighty-eight cases in which firm age was not available from this source, we hand-collect it from IPO prospectuses. We make these data available in Internet Appendix I. For all companies that filed an IPO after April 5, 2012, we check the prospectuses to determine if they filed under the Jumpstart Our Business (JOBS) Act.

For each IPO firm, we seek to download all SEC comment letters related to their IPO. Specifically, we download all "UPLOAD" documents that were

⁵ These data come from the Field-Ritter data set of company founding dates, as used in Field and Karpoff (2002) and Loughran and Ritter (2004). We thank Jay Ritter for making these data publicly available (see <https://site.warrington.ufl.edu/ritter/ipo-data/>).

registered on the company's EDGAR directory up to 2 years after their IPO.⁶ This long window allows for the fact that while SEC letters pertaining to an IPO are written before the company goes public, they are not publicly released and filed on EDGAR until a minimum of 20 business days after an IPO (45 calendar days up until 2012) and sometimes much longer. To determine whether each letter is IPO related, we retain from the letter's heading the SEC file number, which is a unique number that identifies all related filings; we manually check all letters without file numbers. For example, letters relating to draft registration statements (DRS filings, which are the confidential versions of the prospectus that companies release prior to publicly announcing planned IPOs, as permitted under the JOBS Act) generally do not have file numbers and thus need to be manually linked to IPOs. Our final sample consists of 952 companies.

Table 1 and Figure 1 provide descriptive statistics for these 952 companies. Table A1 in the appendix provides full variable descriptions. Company and offer characteristics are similar to those of other IPO samples (see, e.g., Lowry, Michaely, and Volkova 2017) over similar periods. We focus our discussion here on the less commonly used variables.

To characterize the disclosures of a firm prior to SEC intervention, we employ two metrics based on the firm's preliminary prospectus: the total length of the document and a measure of its uncertainty. Total length represents a simple count of the number of English words in the main body of the prospectus, as described in more detail in the appendix. Following Loughran and McDonald (2011, 2013), uncertainty represents the proportion of words in the preliminary prospectus that capture an uncertain tone, as measured by the uncertain, weak and negative word list categories, where these word lists were developed specifically for the finance context. Within our sample, the mean length of the first prospectus is 89,900 words and the mean uncertainty measure is 2.9%. Both the average prospectus length and the average prospectus uncertainty within our sample are similar to Loughran and McDonald (2013), particularly for the latter years of their sample.

To estimate of the likelihood of problems related to a company's financial disclosures, we employ the F-score, which was developed by DeChow et al. (2011) to capture the likelihood of a financial restatement. DeChow et al. (2011) predict Accounting and Auditing Enforcement Releases (AAERs) for a sample of firms based on seven accounting items: accruals, change in receivables, change in inventory, percent of assets that are neither cash nor property plant and equipment, change in cash sales, change in return on assets, and an indicator for whether the firm issued debt or equity. Using their adjusted coefficients and the accounting values for our sample firms, we calculate the F-score for each of

⁶ We use the SEC file number variable from SDC to link our sample with SEC EDGAR (see Lowry, Michaely, and Volkova 2017 for more details). Although we download letters over an extended period after IPOs to ensure we do not miss any IPO-related letters, over 72% of letters are publicly released and filed on EDGAR within 90 days after IPOs, 88% within 6 months, and over 97% are published within 1 year. The small set of companies (less than 5% of the sample) for which we are unable to match any letters are excluded.

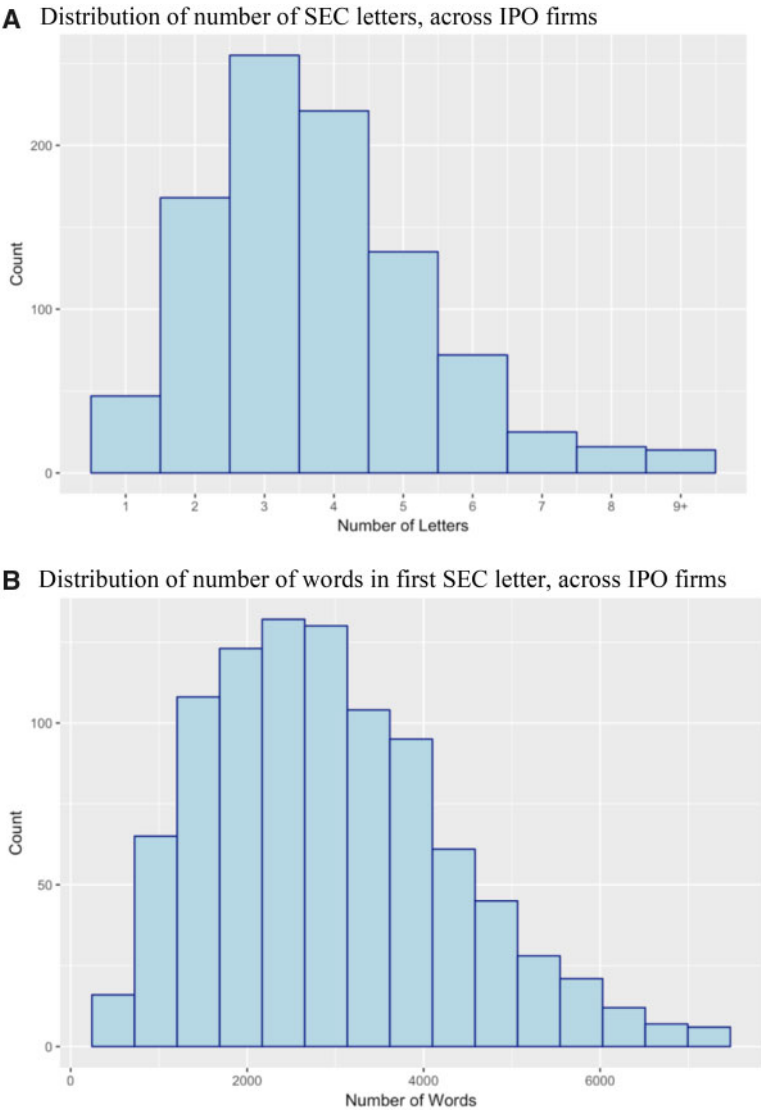


Figure 1
Histograms depicting the SEC’s review process
The sample consists of 952 IPOs between 2005 and 2016, for which we downloaded the SEC’s comment letters from EDGAR. Panel A tabulates the total number of letters each company received from the SEC in response to its IPO prospectus, and panel B tabulates the number of words in the first letter each company receives from the SEC in response to its IPO prospectus.

Table 1
Descriptive statistics*A. Firm characteristics*

Variable	Mean	SD	Pctl(25)	Median	Pctl(75)
<i>Company sales</i>	747.551	4,696.273	24.920	116.514	365.387
<i># of segments</i>	1.462	1.163	1	1	1
<i>Age</i>	19.914	26.109	7	10	19
<i>Underwriter rank</i>	7.867	2.053	7.876	9.001	9.001
<i>Law firm rank</i>	7.961	9.710	1	4	12
<i>VC</i>	0.481	0.500	0	0	1
<i>JOBS Act</i>	0.397	0.490	0	0	1
<i>Length of S-1, words</i>	89,903	25,150	73,320	87,103	103,534
<i>Length of 424, words</i>	96,861	31,927	79,012	92,341	108,186
<i>S1 uncertainty</i>	0.029	0.004	0.026	0.028	0.031
<i>Registration length, days</i>	123.679	129.556	47	93	134
<i>Price update, %</i>	-4.423	20.550	-20.000	0.000	8.164
<i>Shares update, %</i>	0.666	12.666	0.000	0.000	0.000
<i>Size update, %</i>	-3.924	22.892	-20.000	0.000	9.524
<i>F-score</i>	1.187	0.741	0.569	1.071	1.82
<i>Insider sales dummy</i>	0.356	0.479	0	0	1
<i>Volatility, first year</i>	0.581	0.269	0.407	0.535	0.712
<i>Amihud's illiquidity, first month</i>	0.027	0.041	0.003	0.010	0.031
<i>Amihud's illiquidity, first year</i>	0.041	0.056	0.005	0.014	0.049
<i>Secondary shares (dummy)</i>	0.363	0.481	0	0	1
<i>Secondary shares offers, %</i>	14.524	31.748	0.000	0.000	20.000
<i>Overallotment exercised, %</i>	9.646	6.831	0.000	15.000	15.000
<i>Initial returns, %</i>	15.282	25.746	0.000	8.112	23.396
<i>CAR (-2, +2), %</i>	0.074	6.683	-3.147	-0.056	2.415
<i>CAR (+3, +90), %</i>	-0.133	28.020	-17.389	-2.519	12.546
<i>Abnormal return, first half-year, %</i>	1.255	40.576	-23.946	-4.475	19.908
<i>Abnormal return, first year, %</i>	2.543	64.925	-36.840	-7.183	27.189
<i>Nasdaq returns, %</i>	2.405	5.056	-0.551	2.943	5.731
<i>N</i>	952				
<i>B. SEC review characteristics</i>					
<i># letters</i>	3.76	1.71	3.00	4.00	5.00
<i># letters after price range</i>	0.79	1.03	0.00	1.00	1.00
<i># words in the first letter</i>	1966.18	1058.51	1196.00	1851.50	2567.80
<i># questions in the first letter</i>	41.14	20.07	27.00	39.00	53.00

The sample consists of 952 IPOs between 2005 and 2016, for which we downloaded SEC comment letters from EDGAR. Panel A shows descriptive statistics on firm characteristics just before firms' go public, and panel B shows descriptive statistics on the SEC letters. Table A1 in the appendix provides variable descriptions.

our IPO firms.⁷ An F-score of 1.00 indicates that the likelihood of a restatement is equal to the unconditional probability, and higher values represent higher likelihoods. Within our sample, the median (mean) F-score is 1.071 (1.187), with an interquartile range of 0.6–1.8.

Figure 1 describes the SEC letters that pertain to the sample IPOs. Looking first at panel A, one can see that the typical firm receives 3 or 4 letters, but there exists substantial cross-sectional variation, with the number of letters ranging from 1 to 14. The average number of letters per firm varies over time, but it does not exhibit a strong time trend (not tabulated). Panel B of Table 1 shows that the majority of these letters (79%) are received prior to the setting of the

⁷ Because this variable requires 2 years of pre-IPO data, it is only available for 686 of our 952 observations.

initial price range. In fact, 48% of companies do not receive any letters after setting the initial price range.⁸

To maximize the extent to which we can learn about the regulatory process, we take advantage of the fact that SEC letters regarding IPOs have a standard structure. A typical letter has six distinctive parts:

- Part 1: Header – the date of the letter and company information
- Part 2: Title – the reference to the SEC file number of the IPO prospectus
- Part 3: Introduction of the letter – a brief overview of the process, including the next steps the company is required to take
- Part 4: Itemized questions – the main part of the letter, which includes the questions raised by the SEC. The company is required to address these questions in an amended version of the prospectus⁹
- Part 5: Ending – this part provides common instructions on how to reply to the letter
- Part 6: Signature – name of the SEC officer who wrote the letter

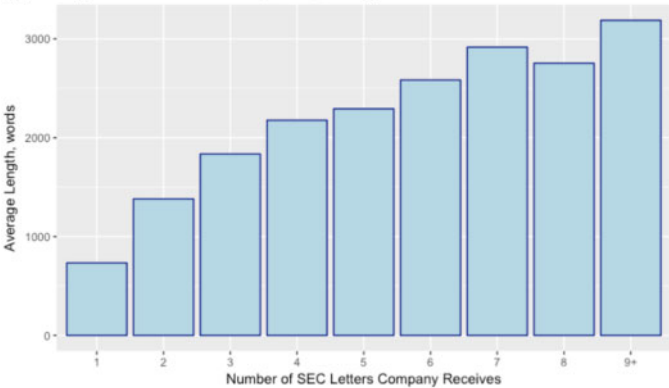
For illustrative purposes, Internet Appendix II shows the first IPO comment letter to Affymax Inc., out of a total of four letters that the company received before it obtained permission from the SEC to list its shares. Our analysis of the substance of SEC concerns focuses on Part 4, the itemized questions. We use the information in this part to develop several empirical measures. Our first measure is the number of words in the first letter a company receives from the SEC. As shown in panel B of Figure 1, most frequently the first comment letter from the SEC has between 1,200 and 2,500 words, but the range extends from less than 100 words to over 15,000 words. In sum, both the number of letters and the length of letters are characterized by high cross-sectional dispersion.

Perhaps not surprisingly, the length of the first letter is positively related to the total number of SEC letters, as shown in panel A of Figure 2. Among companies that receive just one letter, the average length of this letter is 734 words. In contrast, among companies that receive a total of 4 letters, the average length of the first letter is 2,177 words. The correlation between the number of letters and the number of words in the first letter equals 0.51. Panel B shows the evolution of the length of subsequent letters. For instance, the average length of the first letter across all companies is 1,959 words, whereas the second letter is significantly shorter, with only 694 words on average. The decrease in letter length is less pronounced thereafter. Larger companies and companies with longer prospectuses tend to receive longer first letters and more comment letters in general. For instance, companies with sales above 500 million, which

⁸ The timing of when the price range is set has changed over time. Within the subsample of 1996–1997 IPOs examined by Lowry and Schwert (2004), 60% disclosed the price range within the initial prospectus. In comparison, this occurs within only 4.1% of our sample observations.

⁹ For clarity throughout the paper, we refer to the issues raised by the SEC as questions, regardless of whether they are specifically phrased in the form of a question or as a statement.

A Length of the first letters for the companies with a different total number of letters



B Decay in the length of SEC letters

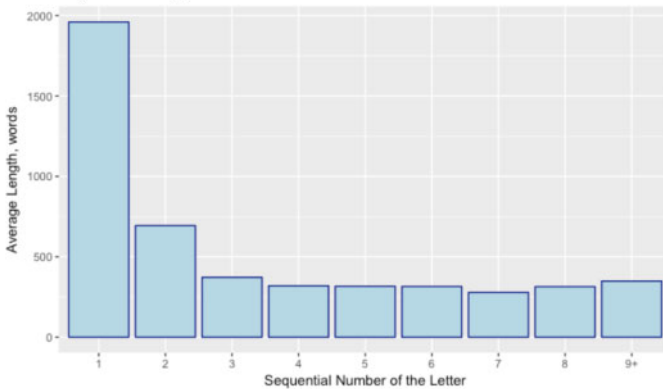


Figure 2
Evolution of the SEC's letter length

The sample consists of 952 IPOs between 2005 and 2016, for which we downloaded the SEC's comment letters from EDGAR. Panel A shows the average length of the first SEC letter, among companies that ultimately received 1, 2,...,9+ SEC letters in total. Panel B shows the average length of the first, second, ..., ninth, or later letter, among the subset of companies that received at least this number of letters.

represent the top 20% of companies, have 2,339 words in the first letter and receive 4.2 letters, on average. In contrast, the bottom 20% companies in sales have 1,410 words in the first letter and receive only 3.2 comment letters, on average.

An alternative metric of the extent of SEC concerns is the number of distinct questions in the letter. Looking at panel B of Table 1, the mean (median) company receives 41 (39) distinct questions from the SEC in its first letter. The SEC's concerns are quite extensive for some companies, as highlighted by the fact that the 75th percentile (maximum) number of SEC questions is 53 (142). While some questions are longer than others and might even contain subbullet points, the correlation between the number of questions and number of

words is 95%, suggesting they capture similar fundamental features of the SEC concerns. Similar to the number of words, the number of questions declines in subsequent letters (not tabulated).

1.2 What does the SEC care about?

There exists potentially meaningful variation in the topics raised by the SEC and in their relative importance, beyond what is captured by the number of letters each company receives and the number of words per letter. For example, the SEC may raise significant concerns about accounting practices for some firms but ask more detailed questions about valuation-related issues for others. The topics of the letters are interesting in their own right, and they are potentially related to the severity of the SEC concerns. To this end, we employ textual analysis techniques to evaluate the specific topics and provide a more complete characterization of the SEC review.

We employ LDA, also sometimes referred to as topic modeling, to determine the topics of SEC focus, and their relative importance. LDA is based on machine learning and has the advantages of being objective and replicable and of producing an output that corresponds to human ability to detect topics (see, e.g., Chang et al. 2009). This algorithm does not require any prior knowledge about the content of the documents; rather, the algorithm identifies the topics. A topic is defined as a group of words that tend to appear in the same context. Words used in conjunction with or in the same neighbourhood of each other will tend to be categorized together. It is a more sophisticated structure than the dictionary “bag of words” approach (e.g., Loughran and McDonald 2011, 2013) because the same word, when used in different contexts, can be classified into different topics. For instance, the meaning of the word “cash” is found within different categories, but it tends to be associated with different words in different cases. The phrase “cash flow” is more likely to refer to financial statements and thus pertain to accounting issues, whereas the phrase “cash bonus” tends to refer to compensation. In sum, the LDA determines the context in which a word is used.

While relatively common within the computer science literature, LDA has only recently began to be used in finance. Nevertheless, it has already proven a powerful technique to uncover fundamental aspects of firms and also of regulatory processes. Wu (2017) investigates firm disclosures to characterize firm-specific shocks to production; Hoberg and Lewis (2017) examine the ways in which 10K disclosures by fraudulent managers differ from other firms; Bellstam, Bhagat, and Cookson (2019) analyze analysts’ reports to characterize firms’ innovativeness; and Ganglmair and Wardlaw (2017) investigate the completeness of loan contracts.

One of the main features of LDA analysis is that it not only provides us with a cluster of topic words, but it also provides an estimate of the importance of each topic in each document. Our main analysis focuses on the first SEC letter firms receive. For each firm, we calculate the number of words in each topic,

equal to the document loading on the topic multiplied by the total number of words in the document. This represents the SEC's focus on a particular topic for a given firm. Hence, LDA allows us to not only find the main concerns but also determine the intensity of each concern in the SEC letters that each firm receives.

While LDA analysis does not require any human judgement in topic detection, it does require one input from the researcher, namely, the optimal number of topics. If the number of topics is too high then some topics might be either duplicative or split overly narrowly into subtopics, but if the number is too low then some key themes might be omitted. To avoid introducing our own priors on the optimal number of topics, we follow two alternative approaches advocated by Goldsmith-Pinkham, Hirtle, and Lucca (2016) and Jegadeesh and Wu (2017), both of which let the data dictate the optimal number of topics. We briefly discuss each of these approaches here, and we refer the interested reader to these papers for more detail.

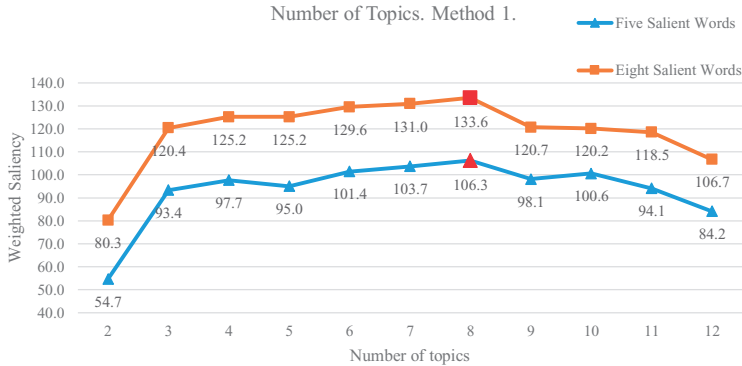
The Goldsmith-Pinkham, Hirtle, and Lucca (2016) approach involves estimating the LDA analysis across varying numbers of topics (e.g., 2, 3,...,12) and determining the weighted saliency of the topics for each case. Following Chuang, Manning, and Heer (2012), the most salient words in a topic are those that are used most frequently within this topic but more rarely in other topics.¹⁰ Figure 3 shows how the weighted saliency varies with the number of topics. Consistent with distinctiveness increasing as the number of topics increases but commonality decreasing, the function has a smooth hump-shaped pattern. The maximum weighted saliency occurs at eight topics, suggesting this is the optimal number of topics across the SEC letters.

A potential weakness of the Goldsmith-Pinkham, Hirtle, and Lucca (2016) approach is that it is sensitive to the choice of which words to exclude (i.e., whether to exclude from the analysis frequent words like pronouns, names of the months, etc.) and whether to use root words or full words. Jegadeesh and Wu (2017) use an alternative method to find the number of topics, which is less sensitive to the above-mentioned issues and which is more widely used in computer science (Manning, Raghavan, and Schütze 2008). Specifically, they employ a hierarchical Dirichlet process (HDP), which is a more advanced version of LDA and enables the machine to determine the number of topics. Because the generated optimal number of topics is mechanically related to the number of documents, we follow Jegadeesh and Wu (2017) and run this process on 5% of the total number of documents in the sample, in our case, on 200 letters.¹¹ As shown in panel B of Figure 3, this method similarly generates

¹⁰ The saliency of the word w in the topic T is defined as $\text{saliency}(w) = P(w) \cdot \sum^P(T|w) \log \frac{P(T|w)}{P(T)}$, where $P(w)$ is the unconditional probability of the word w ; $P(T)$ is the unconditional probability of the topic T ; and $P(T|w)$ is the probability of topic T conditional on the appearance of the word w , which is determined from LDA output using the Bayesian formula.

¹¹ For robustness, we also run the algorithm on repeated subsamples of 100 long documents, defined as documents with more than 1,000 words. This is similar to Jegadeesh and Wu (2017), who also employed samples of 100

A The Goldsmith-Pinkham, Hirtle, and Lucca approach to determining the optimal number of topics



B The Jegadeesh and Wu approach to determining the optimal number of topics

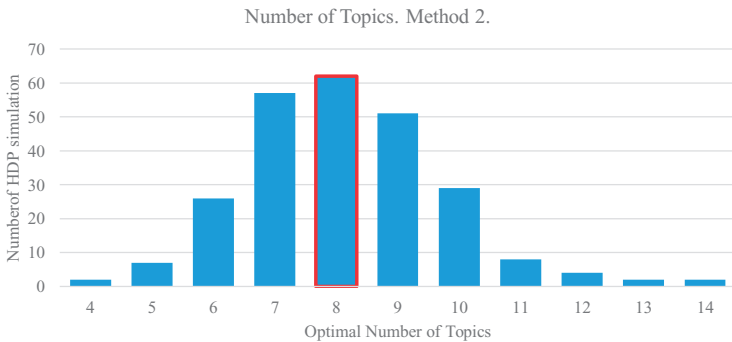


Figure 3
Number of topics for LDA analysis

The sample consists of 952 IPOs between 2005 and 2016, for which we downloaded the SEC comment letters from EDGAR. This figure shows the output from two alternative approaches to detect the optimal number of topics for the LDA analysis. Panel A uses the approach of Goldsmith-Pinkham, Hirtle, and Lucca (2016), in which we define weighted saliency across either the top-five most salient words or the top-eight most salient words in each topic. Under both saliency measures, the criterion function is highest at eight topics, indicating that the LDA analysis should be estimated across eight topics. Panel B uses the approach of Jegadeesh and Wu (2017) with 250 HDP iterations. The panel shows the histogram, which plots the criterion function from this approach, with 2, 3,..., 12 topics. The mean (median) value of the number of topics is 8.132 (8.00), which similarly indicates that the LDA analysis should be estimated across eight topics.

the result that the optimal number of topics is eight. While this method is theoretically grounded, it is not without shortcomings, most importantly that the computing power it requires is substantially greater, particularly for analyses of longer documents.

Having established that eight represents the optimal number of topics within our sample, the next step is to estimate the LDA analysis across our sample

documents and whose sample consists of documents that are on average longer than the SEC letters. Results are qualitatively similar.

of SEC letters. Figure 4 depicts the word clouds. For each of the eight topics, we depict the 100 most salient words. Within each word cloud, the font size represents the relative saliency of each word.

While LDA analysis identifies distinct topics within the letters, the primary economic focus of the topic is not always apparent. Based on the LDA output, there are an average of 250 words per topic in each letter, with many words having the potential to be used in different contexts. In most economic settings, identifying the focus and context of each topic is particularly important. In fact, it is of primary importance in our setting as we seek to analyze the extent to which the SEC's concerns map into several economically meaningful dimensions: the SEC's own stated economic objectives when reviewing the IPO prospectus, company characteristics, disclosure choices in the first preliminary prospectus, and company value.

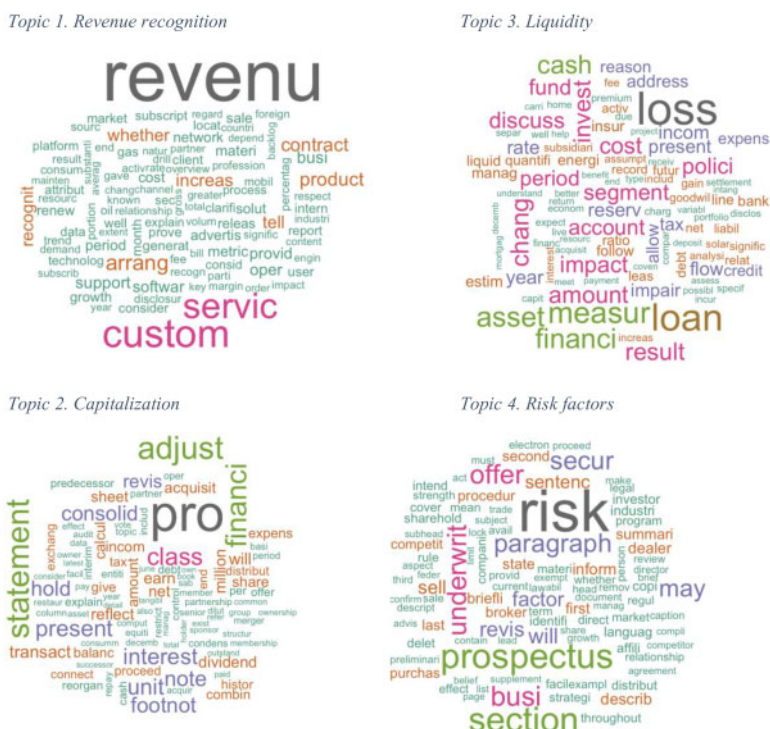


Figure 4
Topic word clouds

The sample consists of 952 IPOs between 2005 and 2016, for which we downloaded the SEC's comment letters from EDGAR. We obtained the first letter that each company received from the SEC in response to its prospectus. We employ topic modeling analysis (LDA) across this set of letters to identify eight topics. This figure illustrates the most salient root words in each topic. The size of the word corresponds to its saliency within the topic. The calculation of word saliency in a topic follows the Chuang, Manning, and Heer (2012) method.



Figure 4
Continued

As a first step to evaluating the economic meaning, we examine the most salient words within each topic. As shown in Table 2 and Figure 4, the most salient root words in the first topic are “revenu,” “custom,” “servic,” etc., whereas the most salient words within the fifth topic are “stock,” “valu,” “fair,” etc. As expected, there is a similarity of words within a topic. However, this information is generally insufficient to unambiguously identify the economic focus of the topic.

To ensure that ambiguity does not affect our conclusions and to maximize the objectivity of the categorization, we rely on the structure of the SEC letters to identify economic meaning in an objective (and replicable) way that does not rely on human input. The SEC structures the letters as sets of paragraphs that fall under specific headings. For example, as shown in the SEC letter provided in Internet Appendix II, the first paragraph falls under the heading “General,” the second paragraph falls under the heading “Management’s Discussion and Analysis of Financial Condition and Results of Operations,” etc. We tabulate the unique headings across our entire sample of SEC letters. Across the total of more than a thousand different item headers, 25 are used in at least 100 different

Table 2
Topics description

A. Eight topics from LDA analysis of SEC letters

Topic number	1	2	3	4	5	6	7	8
Average # of words	278.19	223.05	225.21	363.85	227.17	283.82	246.35	103.65
Median # of words	302.00	221.00	211.00	396.00	231.00	304.00	203.00	104.00
Top-10 root words by saliency	revenu, custom, service, arrang, product, contract, increas, tell, whether, recognit	pro, adjust, statement, financi, class, present, unit, interest, note, footnot	loss, oper, loan, measur, asset, financi, cash, chang, result, invest	risk, prospectus, section, busi, underwrit, offer, secur, paragraph, may will	stock, valu, fair, share, option, common, price, valuat, prefer, grant	compens, execut, item, regul, office, disclosur, award, name, perform, director	product, agreement, develop, clinic, licens, studi, trial, patent, file, expand	respons, prior, note, letter, date, revis, howev, request, believ, unclear
Top-5 companies	Twitter Inc. Zynga Inc. TubeMogul Trulia Inc. Borderfree Inc.	Fifth St Asset Mgmt Inc. Moelis & Co. Camping World Holdings Inc. RCS Capital Corp. .Medley Management Inc.	Approach Resources Inc. Sanchez Energy Corp Antero Resources Corp. Sealy Corp. Springleaf Holdings Inc.	opionsXpress Holdings Inc. CF Industries Holdings Inc. Lincoln Educational Services Las Vegas Sands Corp. Goodman Global Inc.	The Joint Corp. Veraz Networks Inc bluebird bio Inc. Bright Horizons Family Aruba Networks Inc.	Elie Mae Inc. Trans I Inc. Interactive Brokers Group Inc. GAIN Capital Holdings Inc. Zoetis Inc.	Portola Pharmaceut Inc. Ardelyx Inc. Dermira Inc. Immune Design Corp. Zafgen Inc.	Enanta Pharmaceut Inc. SolarCity Corp. Xoom Corp. Ruckus Inc Marrone Bio Innovations Inc

B. Item headers from SEC letters linked to our topics

Item header in SEC letters	Revenue recognition	Capitalization	Liquidity and capital resources	Risk factors	Stock-based compensation	Executive compensation	Intellectual property	Boilerplate
# paragraphs across all company letters	281	192	392	769	175	277	165	n/a
Topic cloud to which this item header is matched	1	2	3	4	5	6	7	8
% of paragraphs matched to this topic cloud	78	46	46	57	91	91	81	n/a

C. KL divergence between SEC letter topics and prospectus topics

KL divergence to closest SI topic	1.00	1.83	1.14	1.11	1.20	1.23	1.00	1.49
Average number of words in the closest SI topic	2,425	2,425	3,192	11,356	5,543	6,690	1,433	2,022
Average number of words in the closest 424 topics	2,528	2,528	3,570	12,063	6,513	6,949	1,571	3,788

The sample consists of 952 IPOs between 2005 and 2016, for which we downloaded SEC comment letters from EDGAR. For each IPO, we analyze the content of the first SEC comment letter with an eight-topic LDA analysis. Panel A reports the average and median number of words in the first comment letter devoted to each topic, the most salient root words in each topic, and the companies that have the highest number of words devoted to each topic. Panel B shows the label of each topic, which is determined as follows. We first extract the twenty-five most common paragraph names across all SEC letters (reported in Table A2 in the appendix). For each topic, we then determine the name of the paragraphs from the SEC comment letters where the topic is most prevalent. For instance, across all the letters a total of 281 paragraphs appear under the title "Revenue Recognition." Seventy-eight percent of these paragraphs have topic number 1 as the most frequent, and hence we label topic 1 as Revenue Recognition. The last topic (topic 8) was not the most frequent in any of the twenty-five sets of paragraphs we tested, and we therefore label it as boilerplate speech. Panel C shows the KL divergence between the SEC letter topic and the closest S-S-1 topic, as well as the number of words in the corresponding

SEC letters. Table A2 in the appendix lists these twenty-five headers, along with the percentage of companies' letters that include this heading and the number of paragraphs across the texts of all letters that fall under the heading.

For each of these twenty-five-item headings, we apply the previously trained LDA model (with eight topics) on all paragraphs that fall under the heading. Panel B of Table 2 shows the heading that is matched most closely to each of our eight topics. Looking at Column 1, one sees a total of 281 paragraphs that fall under the Revenue Recognition heading across our sample of SEC letters; in 78% of these paragraphs, Topic 1 is the most heavily represented. Intuitively, the concentration of words in the paragraphs that fall under the Revenue Recognition item heading matches most closely to the concentration of words in the Topic 1 word cloud. As shown in Table 2 and Figure 4, this approach labels the other seven topics as "Capitalization," "Liquidity," "Risk Factors," "Stock Compensation," "Executive Compensation," "Intellectual Property," and "Boilerplate."

We conduct a detailed verification analysis of these labeling. Across the 200 most common words in each of our eight LDA topics, we ascertain (1) whether the word is among the most common words in the matched topic (i.e., the paragraphs falling under the matched section header within the letters), and (2) in cases in which it is not among the most common words in the matched topic – within which topic is it the most common. As shown through a series of figures in Internet Appendix III, the vast majority of the most common words in each topic are also among the most common words in the paragraphs that fall under the matched item headers. Importantly, the remaining words come from a wide variety of other item headers, with results suggesting no concentration of words within a "second most common item header." (Internet Appendix III provides a more detailed description.)

The labels deserve further comment on two topics. First, the intellectual property topic is heavily concentrated within the pharmaceutical industry, a finding that is consistent with intellectual property-related matters being particularly important for this set of companies. As shown in panel A of Table 2, across all other topics the companies represent a variety of industries. Second, the one topic for which we obtain no close match is Topic 8. There is no concentration between this topic's word cloud and any of the twenty-five paragraph headings. Indeed, examining the content of words and phrases in this topic suggests it mostly comprises boilerplate words, such as date, letter, and requests, and we therefore label it as such.

The method we develop to determine the economic meaning of LDA-generated topics is objective and does not require human input. As such, it substantially increases the applications as well as the informativeness of text-based analysis. The procedure, which is based on a paragraph labeling approach, is relevant in many other settings. Many financial documents, for example, 10Ks, S-1's, and proxies, have a standard structure that enables an analogous approach. In our context, a main takeaway is that the topics of focus in the

SEC letters clearly relate to the components of disclosure required to register securities sold in the United States: “(i) a description of the company’s business; (ii) a description of the security to be offered for sale, (iii) information about the management of the company; and, (iv) financial statements certified by independent accountants.”¹² This is consistent with the SEC concentrating to a large extent on its mandate when it comes to the IPO review process, and with our topic modeling capturing this process.

1.3 Relating topics in SEC letters to topics in prospectuses using KL divergence

In addition to analyzing the SEC letters themselves, we also consider their information content relative to the voluntary disclosure in the prospectus. Because IPO firms have little public history, the prospectus represents a strong proxy for the information environment surrounding the firm. As described in this section, we borrow and combine (for the first time to the best of our knowledge) two methods from computer science (LDA and KL divergence) to isolate the marginal information content that can be attributed to the interaction between the SEC and the IPO firm, as characterized through the SEC letters.

We use both topic models analysis (on the prospectus and SEC letters separately) and a technique developed by Kullback and Leibler (1951), KL divergence, to identify the letters’ marginal information value. The first step is to apply the LDA topic modeling to the preliminary prospectus of each company, analogous to the previously described analysis of the SEC letters. The twenty-eight prospectus topic clouds are shown in Internet Appendix IV (where the choice of twenty-eight is based on the Jegadeesh and Wu method, as described previously). The greater number of topics in the prospectus (vs. the SEC letters) is consistent with the fact that the prospectus is much longer. Intentionally, this process is conducted completely independently of the SEC letter classification so that we do not a priori impose any relationship between the topics in the prospectus and the letters. Thus, the prospectus and letter topics could be completely unrelated, or they could be very closely related.

The second step is to quantify the relationship between the SEC letter topics and the prospectus topics. We employ KL divergence as a metric to achieve this. Recognizing that all topics in both the SEC letters and the prospectuses can be viewed as probabilistic distributions spanned over the dictionary of all words, we apply ideas from information theory to the extracted topics. Specifically, we employ statistical methods developed in Burnham and Anderson (2004), and use KL divergence to identify the SEC letter topic with the least incremental information relative to each prospectus topic. For each of the eight SEC letter topics (characterized as 8 vectors), we calculate the KL divergence with each

¹² <https://www.sec.gov/answers/regis33.htm>

of the 28 prospectus topics (analogously characterized as 28 vectors).¹³ Thus, we form an 8 x 28 matrix of KL divergence measures, where the KL metric represents a measure of the incremental information in the SEC topic relative to the respective prospectus topic. The final step is to select, for each SEC letter topic, the prospectus topic that is most similar, that is, the prospectus topic with the smallest KL distance. For each SEC letter topic, panel C of Table 2 shows the KL divergence to the closest prospectus topic, as well as the average number of words in the matched topic within both the preliminary prospectus (S1) and final prospectus (424). Across all topics, we see that company disclosure increases. The next section examines more directly the relation between these increases and SEC questions on the topic.

Importantly, KL divergence satisfies three key criteria: it is replicable and does not rely on human judgment, it puts a higher emphasis on more frequent words, and it is robust to the exclusion of low frequency words within the topic. In contrast, cosine similarity, which has been widely used to estimate the distance between two vectors of word frequencies, meets the first requirement, partly addresses the second, and does not satisfy the last. Replications of the topic matchings illustrate these features. Specifically, we select the most frequent words in each of the SEC letter topics and in each of the prospectus topics, where the number of words selected varies from 1,000, 1,100, 1,200,...,4,000. Across all cases, the KL divergence matching stays the same, whereas the cosine similarity matching is sensitive to such choices.

KL divergence outputs enable us to link changes in companies' disclosures to the topics of SEC concerns so as to examine, for example, the extent to which companies increase accounting-related disclosures in response to accounting-related questions from the SEC. In addition, this process also enables us to control for companies' initial disclosures on each topic.

Table 3 shows regressions of attention to each topic, on company and offer-specific characteristics as well as the extent of initial disclosures on the topic. Results indicate that the intensity of the SEC's questions is positively related to company information asymmetry. The only company characteristic that we find systematically related to specific SEC topics is the company's accounting risk, which we measure with the F-Score. The F-score is significantly positively related to SEC comments on both revenue recognition and capitalization, but it is not significantly related to any other topics. Subsequent sections examine the extent to which SEC concerns on these topics are related to company dynamics both at the time of IPOs and over the subsequent year.

¹³ The length of the vector is equal to the total number of potential words (defined as words appearing in any company's prospectus and/or SEC letter). Each component of the vector represents the weight of the corresponding word in the topic cloud.

Table 3
Determinants of topics in the SEC's comment letters

Topic	Dependent variable: <i>Number of words in the corresponding topic in the first SEC letter</i>							
	<i>Revenue recognition</i>	<i>Capital-ization</i>	<i>Liquidity</i>	<i>Risk factors</i>	<i>Stock compens</i>	<i>Executive compens</i>	<i>Intellectual property</i>	<i>Boiler</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Information asymmetry								
<i>Age</i>	-0.112*** (0.041)	-0.105** (0.041)	-0.054 (0.047)	-0.042 (0.038)	-0.109*** (0.041)	-0.006 (0.028)	-0.078* (0.044)	-0.025 (0.024)
<i># of segments</i>	-0.019 (0.029)	0.020 (0.034)	0.045 (0.040)	-0.0003 (0.026)	-0.007 (0.026)	-0.003 (0.025)	-0.013 (0.034)	0.041 (0.029)
Advisor quality								
<i>UW rank</i>	0.031 (0.021)	0.020 (0.016)	0.012 (0.027)	0.004 (0.017)	0.039** (0.017)	0.059*** (0.016)	-0.016 (0.023)	-0.025** (0.012)
<i>Law firm rank</i>	0.001 (0.003)	-0.006* (0.003)	0.001 (0.003)	-0.003 (0.003)	-0.001 (0.004)	-0.006 (0.004)	0.002 (0.003)	-0.001 (0.002)
Company disclosure								
<i>S1 length</i>	-0.591*** (0.154)	0.199 (0.174)	0.180 (0.156)	-0.487*** (0.147)	-0.169 (0.186)	-0.386*** (0.099)	0.265 (0.182)	-0.094 (0.075)
<i>S1 uncertainty</i>	0.212 (13.766)	-38.346*** (8.888)	11.332 (14.337)	6.519 (13.313)	-5.033 (10.514)	8.319 (7.574)	20.458*** (7.677)	-4.957 (8.252)
Company accounting risk								
<i>High F-score</i>	0.134*** (0.040)	0.128** (0.050)	-0.011 (0.042)	0.019 (0.042)	0.058 (0.056)	-0.005 (0.064)	0.062 (0.046)	0.042 (0.037)
Control variables								
<i>Company size</i>	0.096*** (0.028)	0.072*** (0.025)	0.051* (0.027)	0.056*** (0.013)	0.009 (0.017)	0.025 (0.021)	-0.071* (0.043)	0.032* (0.018)
<i>VC</i>	0.160 (0.097)	-0.571*** (0.097)	-0.335*** (0.058)	-0.200*** (0.076)	0.189** (0.077)	-0.154** (0.070)	0.093* (0.056)	0.041 (0.049)
<i>JOBS Act</i>	-0.081 (0.155)	-0.137 (0.171)	-0.308*** (0.088)	-0.048 (0.072)	-0.230 (0.161)	-0.471*** (0.102)	-0.038 (0.070)	-0.073 (0.100)
<i>Corresponding topic in S1</i>	-0.003 (0.028)	0.012 (0.018)	0.035 (0.026)	-0.001 (0.043)	0.024 (0.022)	0.008 (0.012)	0.009 (0.034)	0.004 (0.008)
<i>Industry, year FEs</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	686	686	686	686	686	686	686	686
<i>R²</i>	.592	.546	.584	.498	.481	.471	.654	.327

The sample consists of 686 IPOs between 2005 and 2016, for which we download SEC comment letters from EDGAR and that have at least 2 years of Compustat data pre-IPO. Eight topics are extracted from the first SEC comment letters of each company, using LDA analysis. This table shows the relation between SEC focus on a topic and company and offer characteristics. Each topic is measured as the natural logarithm of (number of words in the topic), where the number of words in the topic equals length of the letter * the percentage of letter dedicated to each topic (i.e., LDA output). Independent variables are divided into five groups: company information uncertainty (company age and number of segments), advisor quality (underwriter rank and law firm rank), company disclosure (S1 length and S1 uncertainty), company accounting risk (F-score) and other control variables (company size, VC backing, JOBS Act, and length of corresponding topic in S1). The corresponding topic in the S1 is determined by KL divergence, where we match the 8 SEC letter topics to the 28 S1 prospectus topics. All regressions include Fama-French 48 industry and year fixed effects. All errors are robust and clustered at the Fama-French 48 industry level. Table A1 in the appendix provides variable descriptions.

2. Information Content and Consequences of the SEC Pre-IPO Process

2.1 SEC review and information revealed around IPOs

Here, our objective is to determine whether the SEC review results in greater information revelation by companies—consistent with the SEC’s mission of protecting investors—and also whether it leads to delays in the going public process, which can be costly for several reasons. First, delays in going public expose firms to the risk of changes in market conditions and might increase the probability of IPO withdrawal. Second, management is distracted from the

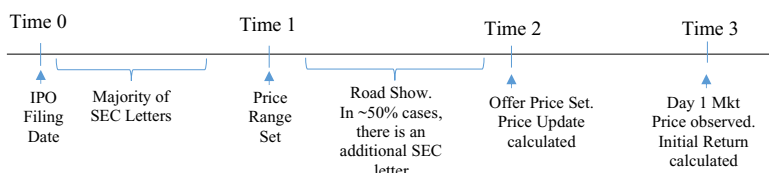
Table 4
Are SEC comment letters related to IPO offer characteristics?

	<i>Registration period length</i>	<i>Change in disclosure</i>	<i>Price update</i>	<i>Share update</i>	<i>Size update</i>
# SEC letters	0.661*** (0.061)	0.066*** (0.016)			
# SEC letters (before price range)			-6.658*** (1.777)	-2.026* (1.063)	-8.077*** (1.986)
Company size	0.001 (0.016)	0.001 (0.003)	1.726*** (0.482)	0.272 (0.570)	2.010*** (0.777)
# of segments	0.024 (0.017)	0.005 (0.004)	-0.801 (0.522)	-0.242 (0.306)	-0.860 (0.605)
Age	0.019 (0.017)	-0.007 (0.006)	-1.669 (1.030)	0.855 (0.722)	-1.215** (0.537)
Underwriter rank	-0.056*** (0.013)	0.003 (0.003)	0.538 (0.760)	0.300 (0.592)	0.868 (1.113)
Law firm rank	-0.006*** (0.002)	0.0001 (0.0004)	0.135* (0.071)	0.039 (0.045)	0.152* (0.082)
VC	0.012 (0.045)	-0.037*** (0.009)	6.646*** (2.285)	2.739** (1.096)	9.291*** (2.025)
JOBS	-1.022*** (0.087)	-0.051*** (0.013)	2.846 (2.898)	1.090 (1.933)	3.590 (4.513)
S1 length	-0.083 (0.078)	-0.026 (0.020)	-6.941* (3.941)	0.798 (2.829)	-6.962 (5.778)
S1 uncertainty	0.710 (5.955)	4.936*** (1.023)	31.790 (278.230)	-221.275** (100.881)	-280.315 (320.611)
Industry, year FEs	Yes	Yes	Yes	Yes	Yes
Observations	952	952	952	952	952
R ²	.597	.286	.203	.078	.182

The sample consists of 952 IPOs between 2005 and 2016, for which we download SEC comment letters from EDGAR. The dependent variable in Column 1 is registration period length, which is measured as the natural logarithm of the number of days between filing date of the first public version of the prospectus and the IPO date. The dependent variable in Column 2 is change in disclosure, measured as $\ln\left(\frac{\text{words}_{424}}{\text{words}_{S1}}\right)$, where the 424 is the final version of the prospectus and the S1 is the first version of the prospectus. The dependent variable in Column 3 is price update, measured as the percentage change from the midpoint of the initial price range to the IPO offer price. The dependent variable in Column 4 is share update, measured as the percentage change from the shares filed to shares offered. The dependent variable in Column 5 is size update, measured as the percentage change from the initial filing amount (equal to shares filed times midpoint of filing range) to final offer amount (equal to shares offered times offer price). All regressions include Fama-French 48 industry and year fixed effects, and errors are robust and clustered at the Fama-French 48 industry level. Table A1 in the appendix defines the variables in detail.

core business. Third, many firms depend on proceeds from their IPO to finance their investment, and delays can result in projects not taken.

Regressions in Columns 1 and 2 of Table 4 focus on these issues, where the independent variable of interest is the number of SEC letters. The dependent variable in Column 1 is the length of the registration period, defined as the log of number of days between the preliminary prospectus filing date and the IPO date. In Column 2, the dependent variable is the extent of the changes in the prospectus following the SEC's comments, defined as the log of number of words in the final prospectus relative to number of words in the preliminary prospectus. Several interesting observations emerge. First, registration period length is significantly positively related to the number of rounds of review with the SEC. One additional letter from the SEC is associated with a 15-day longer

**Figure 5****Timing of information production in our paper versus that in the prior literature**

This figure shows the typical time line of an IPO for offerings during our 2005–2016 sample period.

registration period.¹⁴ Second, Column 2 indicates that a greater number of rounds with the SEC is associated with greater increases in company disclosure: one more SEC letter from the median value is associated with a 1.5% increase in prospectus length.

The SEC review process also potentially relates to the dynamics of the IPO pricing process. As shown in Figure 5, which illustrates the typical time line, most communication with the SEC occurs prior to setting the initial price range: 79% of SEC letters occur before the setting of the initial price range, and 48% of companies receive no letters after they set their price range (as previously shown in panel B of Table 1). Thus, the underwriter and the firm's management know almost all SEC concerns when the price range is set. If underwriters fully incorporate this information into the initial price range, then the number of SEC letters should not be related to the price update, defined as the percentage change between this initial price range and the final offer price.

However, Lowry and Schwert (2004) find that underwriters do not fully incorporate all public information into the initial price range. In a similar vein, Cornelli and Goldreich (2003) and Loughran and Ritter (2004) find that underwriters do not fully incorporate all public information at the next step in the IPO process, that is, when they set the offer price. If underwriters only partially incorporate SEC concerns into the price range and institutional investors express similar levels of concern during the roadshow (either from their own research and/or by extrapolating information from the number of prospectus revisions), then the underwriter will be compelled to lower the offer price. This will result in a negative relation between SEC concerns and the price update. Analogously, in a scenario in which the SEC had fewer concerns than anticipated, underwriters might be hesitant to incorporate such optimism into the price range until institutional investors similarly express a low level of concern, again resulting in a negative relation between the extent of SEC concerns and the price update.

Consistent with this conjecture, Column 3 of Table 4 shows that the number of SEC letters is significantly negatively related to the price update. We measure

¹⁴ The estimate is taken for a median company, which receives four letters and has a registration length of 93 days. Additional delay from one extra letter would be associated with $\exp[0.661 \cdot \log(5/4) + \log(93)] - 93 = 15$ days.

Table 5
Are the SEC's comment letters related to firm uncertainty after their IPO?

	<i>Initial returns</i>	<i>Volatility, first year</i>	<i>Amihud's illiquidity, first month</i>	<i>Amihud's illiquidity, first year</i>	<i>Insider sales</i>	<i>Abnormal return, first year</i>
# SEC letters	-1.639 (2.104)	0.089*** (0.021)	0.010*** (0.003)	0.007** (0.003)	0.081** (0.036)	3.379 (6.681)
Company size	0.762 (0.564)	-0.014*** (0.003)	-0.006*** (0.002)	-0.011*** (0.003)	0.016 (0.012)	2.488 (1.791)
# of segments	-0.579 (0.575)	-0.005 (0.004)	0.001 (0.001)	0.001 (0.001)	-0.005 (0.020)	-0.215 (1.583)
Age	-0.162 (1.052)	-0.011** (0.006)	0.0001 (0.001)	0.002* (0.001)	-0.004 (0.019)	3.275* (1.989)
Underwriter rank	0.922 (0.828)	-0.008 (0.006)	-0.006** (0.003)	-0.010*** (0.003)	0.019*** (0.007)	2.318 (1.618)
Law firm rank	0.190* (0.099)	0.0002 (0.001)	-0.0002** (0.0001)	-0.0004*** (0.0001)	-0.002 (0.002)	0.242 (0.227)
VC	12.116*** (2.713)	0.053*** (0.015)	-0.006* (0.003)	-0.012** (0.005)	0.137** (0.066)	-6.418 (5.756)
JOBS Act	3.610 (3.019)	0.055*** (0.015)	-0.008*** (0.003)	-0.011*** (0.004)	-0.036 (0.068)	4.438 (7.755)
SI length	-9.846*** (3.345)	-0.001 (0.024)	-0.012 (0.010)	-0.012 (0.010)	0.136** (0.066)	0.965 (8.476)
SI uncertainty	-461.760** (216.723)	2.083 (1.574)	-0.005 (0.536)	-0.418 (0.559)	0.634 (7.379)	841.060 (793.770)
Industry, year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	952	952	952	952	952	952
R ²	.168	.451	.396	.507	.133	.093

The sample consists of 952 IPOs between 2005 and 2016, for which we download SEC comment letters from EDGAR. The dependent variable is initial returns in Column 1, volatility (the standard deviation of daily returns over the 1 year following the IPO) in Column 2, Amihud's illiquidity measured over the first month after the IPO in Column 3, Amihud's illiquidity measured over the first year after the IPO in Column 4, insider sales at the end of the lockup period (a dummy variable equal to 1 if there were any insider sales between 170 and 190 days after the IPO) in Column 5, and abnormal returns over the 1 year after the IPO in Column 6. All regressions are ordinary least squares (OLS), and independent variables include the logarithm of the number of the SEC comment letters, plus company controls and characteristics of the initial disclosure. All regressions include Fama-French 48 industry and year fixed effects, and errors are robust and clustered at the Fama-French 48 industry level. Table A1 in the appendix defines the variables in detail.

the number of letters issued prior to the setting of the initial price range, and we find that one more SEC letter to a median company is associated with a 1.9% lower price update. While investor demand expressed during the bookbuilding process most often causes companies to revise the offer price, sometimes they also revise the number of shares offered. Within our sample, price is revised in 84% of cases, compared to only 16% of cases for shares. Despite this lower frequency, Column 4 shows that greater SEC concerns are similarly negatively related to the share update (defined as the percentage change from shares initially filed to shares offered). Finally, Column 5 shows that greater SEC concerns are associated with a significantly lower total offer size update (defined as the percentage change from {shares filed * midpoint of filing range} to {shares offered * offer price}). In sum, our results are consistent with information uncovered by the regulator being significantly related to estimates of firm value prior to IPOs.

Table 5 examines the extent to which these SEC concerns are related to information asymmetry around the time of IPOs and uncertainty, liquidity,

insider sales, and performance after IPOs. Column 1 focuses on initial returns: if more substantial SEC concerns contribute to uncertainty regarding aggregate market demand for the company, that is, about external investors' estimates of true firm value, then the underwriter will underprice the offering by a greater amount (Rock 1986; Beatty and Ritter 1986). The coefficient on initial returns is insignificantly negative suggesting underwriters do not underprice IPOs more when the SEC expresses greater concerns.

We find however, strong evidence that greater SEC concerns are significantly related to uncertainty that is revealed over the months following IPOs. Looking first at Column 2, one can see that the number of SEC letters is significantly positively related to post-IPO volatility, measured as the standard deviation of daily returns over the 1 year after IPOs. This is consistent with concerns uncovered by the SEC being revealed over time, in ways that relate to uncertainty among investors regarding firm value. Columns 3 and 4 show that greater SEC concerns are also associated with lower post-IPO liquidity; one additional SEC letter to a median company is associated with a 5.4% standard deviation increase in the Amihud illiquidity measure over the 1 month after IPOs, and an analogous 2.7% over 1 year. Conclusions are similar when we use the bid-ask spread as a measure of illiquidity (not tabulated). Looking at Column 5, the significant positive relation with post-IPO insider sales (measured as a dummy variable for whether insiders sell shares in the days around the lockup expiration) provides further corroborating evidence: because insiders typically hold nondiversified portfolios (e.g., Faccio et al. 2011; Lyandres et al. 2019), higher volatility makes their portfolios riskier, and, hence, the incentive to sell (and diversify) is more intense. In economic terms, one more SEC letter to a median company is associated with a 2% higher probability of insider sales and with volatility that is 3.7% higher than the median value. These findings provide added evidence that the severity of SEC concerns represents an indicator of problems within the company.

Finally, Column 6 examines whether SEC concerns predict the direction of post-IPO performance, specifically whether they relate to lower abnormal returns over the first year following IPOs. While the coefficient is negative, it is not significant at conventional levels. We examine this relation in more depth in the next section, where we delve into the specific topics of focus within each company's SEC letters.

2.2 Topics of SEC concern and company responses

After examining the how the extent of SEC concerns (proxied by number of letters) is related to firm dynamics, we now examine the specific topics of SEC concern. Looking first at Table 6, we expect that more SEC scrutiny on any topic will result in increased firm disclosure on that issue within the prospectus. The KL divergence method of matching SEC topics to prospectus topics, as described in Section 1.3, enables us to directly examine this conjecture. Table 6 compares the extent of SEC questions on each topic to the extent of prospectus

Table 6
Firms' responses to SEC inquiries by topic

A. Top-one matched topic

	Dependent variable: <i>Change in disclosure, top KL-matched topic</i>							
	<i>Revenue recognition</i>	<i>Capitalization</i>	<i>Liquidity</i>	<i>Risk factors</i>	<i>Stock compens</i>	<i>Executive compens</i>	<i>Intellectual property</i>	<i>Boiler</i>
<i>Corresponding topic in first SEC letter</i>	0.016** (0.007)	0.031*** (0.010)	0.023** (0.009)	0.016*** (0.005)	0.023** (0.011)	0.008 (0.006)	0.030*** (0.011)	0.011 (0.011)
<i>Company size</i>	0.001 (0.005)	0.0002 (0.004)	0.001 (0.005)	0.003 (0.006)	0.003 (0.005)	0.001 (0.003)	0.006 (0.006)	0.0005 (0.003)
<i># of segments</i>	0.007 (0.004)	0.006 (0.004)	0.005 (0.004)	0.007 (0.004)	0.007 (0.004)	0.006 (0.004)	0.008* (0.004)	0.005 (0.004)
<i>Age</i>	-0.007 (0.006)	-0.005 (0.006)	-0.007 (0.006)	-0.006 (0.006)	-0.006 (0.006)	-0.008 (0.006)	-0.005 (0.006)	-0.008 (0.006)
<i>Underwriter rank</i>	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.004)	0.003 (0.003)	0.004 (0.003)	0.003 (0.003)	0.004 (0.003)
<i>Law firm rank</i>	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.0001 (0.0005)	0.001 (0.001)	0.0001 (0.001)
<i>VC</i>	-0.052*** (0.012)	-0.034*** (0.009)	-0.042*** (0.011)	-0.047*** (0.014)	-0.054*** (0.012)	-0.037*** (0.009)	-0.054*** (0.015)	-0.038*** (0.009)
<i>JOBS</i>	-0.052*** (0.016)	-0.050*** (0.014)	-0.046** (0.018)	-0.051*** (0.017)	-0.049*** (0.016)	-0.051*** (0.014)	-0.054*** (0.017)	-0.053*** (0.014)
<i>S1 length</i>	-0.039 (0.031)	-0.055 (0.034)	-0.056* (0.033)	-0.053 (0.037)	-0.045 (0.031)	-0.024 (0.019)	-0.061 (0.039)	-0.024 (0.019)
<i>S1 uncertainty</i>	5.505*** (1.497)	6.600*** (1.788)	5.042*** (1.414)	5.590*** (1.783)	5.516*** (1.573)	4.531*** (1.249)	5.207*** (1.570)	4.510*** (1.267)
<i>Industry, year FEs</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	952	952	952	952	952	952	952	952
<i>R²</i>	.130	.141	.134	.091	.135	.273	.096	.273

B. Top-three matched topics

	Dependent variable: <i>Change in disclosure, top-3 KL-matched topics</i>							
	<i>Revenue recognition</i>	<i>Capitalization</i>	<i>Liquidity</i>	<i>Risk factors</i>	<i>Stock compens</i>	<i>Executive compens</i>	<i>Intellectual property</i>	<i>Boiler</i>
<i>Corresponding topic in first SEC letter</i>	0.007** (0.003)	0.014*** (0.004)	0.010** (0.004)	0.009*** (0.003)	0.009*** (0.003)	0.004 (0.004)	0.011*** (0.003)	0.007 (0.007)
<i>R²</i>	.235	.249	.239	.237	.238	.232	.240	.232

C. Bottom-three matched topics

	Dependent variable: <i>Change in prospectus, bottom-3 KL-matched topics</i>							
	<i>Revenue recognition</i>	<i>Capitalization</i>	<i>Liquidity</i>	<i>Risk factors</i>	<i>Stock compens</i>	<i>Executive compens</i>	<i>Intellectual property</i>	<i>Boiler</i>
<i>Corresponding topic in first SEC letter</i>	0.001 (0.012)	-0.010 (0.011)	0.001 (0.012)	0.004 (0.008)	0.019*** (0.007)	0.007 (0.011)	0.012 (0.014)	0.011 (0.027)
<i>R²</i>	.143	.115	.143	.136	.222	.123	.118	.373

The sample consists of 952 IPOs between 2005 and 2016, for which we download SEC comment letters from EDGAR. Using LDA analysis, eight topics are extracted from the first SEC comment letters of each company, and twenty-eight topics are extracted from initial S-1 prospectuses. To find the topic(s) in the prospectus that most closely match each of the eight SEC letter topics, we employ KL divergence, as described in more detail in the text. Panel A (panel B) focuses on change in disclosure in the one topic (three topics) that match most closely to the SEC letter topic. Panel C focuses on the change in disclosure on the three topics that match least closely to the SEC letter topic. We apply the trained LDA model from the first prospectuses (S1) to the final prospectuses (424) to determine the word increase in each of these topics, such that the change in disclosure among each of the prospectus topics is measured as $\ln(\% \text{ topic in 424 number of word in 424} / \% \text{ topic in S1 number of word in S1})$. Each column shows an OLS regression where the dependent variable is the change in disclosure in 1 of the 8 topics, and independent variables include the natural log of the number of words in the corresponding topic in the first SEC letter, plus company and offer characteristics. All regressions include Fama-French 48 industry and year fixed effects, and errors are robust and clustered at the Fama-French 48 industry level. Table A1 in the appendix defines the variables in detail.

revision on the single most closely matched topic (panel A) and across the three most closely matched topics (panel B).¹⁵ We regress the percentage increase in words within a specific topic (i.e., between the initial S1 and the final 424) on the extent of questions on that topic (measured as the number of words on this topic in the first SEC letter). As shown in Column 1 of panel A, 1% greater SEC attention to revenue recognition is associated with a 2% increase in attention to this topic within the company's prospectus. Economic magnitudes are similarly meaningful across most of the other topics. We note that the boilerplate provides a good placebo: there is no reason to expect that more words within SEC letters relating to boilerplate issues would relate to a greater increase in boilerplate issues within the prospectus. Consistent with this, we do not find a significant effect here.¹⁶

Panel B of Table 6 shows that results are similar when we compare SEC questions on a topic to prospectus revisions across the three most closely matched topics. The magnitude of the coefficients, however, is lower in all cases, suggesting that this broader match captures more information that is less closely related to the SEC topic.

To further validate the procedure, we use the following falsification test, reported in panel C. We compare SEC questions on a topic to prospectus revisions on the three *least* closely matched topics. Consistent with the KL distance approach meaningfully matching topics across two different types of documents and with companies revising disclosure to the specific SEC questions, we find little evidence of a significant relation in this falsification test: only one of the eight topics is statistically significant. To evaluate the significance of the difference between the relations in panels A and C, we employ a block bootstrap approach (see, e.g., Cameron, Gelbach, and Miller 2008). Similar to conventional bootstrapping, this enables us to generate a distribution of coefficients and thereby assess the significance of the differences, but it offers the additional advantage of maintaining the clustering structure (in our setting at the industry level).¹⁷ Results indicate significantly greater increases in disclosure on a prospectus topic in response to the best-matched

¹⁵ Our focus on the three most closely matched topics is based on a statistical analysis, which indicates that any more distant topics represent significantly worse matches. More formally, for each SEC letter topic, we compare the KL divergence of the first-best matched topic with the KL divergence of each of the other twenty-seven prospectus topics. We compute a pseudo-*t*-statistic as $\frac{(KL(1) - KL(i))}{St.Dev(KL)/\sqrt{28}}$, where the standard deviation is computed across the KL's between this SEC letter topic and each of the twenty-eight prospectus topics. For the second- and third-best matched topics, we fail to find significant differences. In contrast, for the fourth-best matched topic, this pseudo-*t*-statistic indicates significance at the 1% level for 7 of the 8 topics.

¹⁶ Although we focus here on just the first letter, an analysis including all letters yields similar results. However, because using all letters results in an uneven number of letters across IPO firms, we present only the results based on the first letter.

¹⁷ Blocks are defined at the Fama-French 48 industry level. In each step, we randomly (with repeat) select blocks. Then we perform the regression on the pulled observations that belong to the selected blocks. We repeat this analysis 1,000 times and record the coefficients. Significance tests are based on the distribution of these coefficients.

Table 7
Price updates and the content of the first SEC comment letter

Dependent variable: <i>Price update</i>					Obs.
	<i>Topic in the SEC letters</i>	<i>Corresponding topic in S-1</i>	<i>#SEC letters before price range</i>	R^2	
Revenue recognition	-2.504*** (0.863)	1.295*** (0.465)	-5.003*** (1.918)	.216	952
Capitalization	-1.873** (0.906)	1.317*** (0.456)	-5.185*** (1.960)	.214	952
Liquidity	-3.128*** (1.016)	1.390*** (0.485)	-4.532** (1.820)	.219	952
Risk factors	-1.031 (0.939)	0.257 (0.376)	-6.131*** (1.806)	.204	952
Stock compensation	-0.525 (0.711)	1.327*** (0.470)	-6.025*** (1.926)	.211	952
Executive compensation	-0.852 (0.687)	-0.002 (0.174)	-6.310*** (1.786)	.204	952
Intellectual property	-1.213 (0.965)	0.256 (0.379)	-6.196*** (1.741)	.205	952
Boiler	-0.006 (0.902)	-0.012 (0.174)	-6.654*** (1.802)	.203	952

The sample consists of 952 IPOs between 2005 and 2016, for which we download SEC comment letters from EDGAR. Each row shows a regression of price update (equal to the percentage change from the midpoint of the initial price range to the IPO offer price) on the natural log of the number of words in the topic in the first SEC letter, the natural log of the number of words in the topic in the S-1, the number of SEC letters before the price range is set, and company and offer controls (company size, number of segments, age, underwriter rank, law firm rank, S-1 length, and S-1 uncertainty), where coefficients on these controls are suppressed. All regressions include Fama-French 48 industry and year fixed effects, and errors are robust and clustered at the Fama-French 48 industry level.

SEC letter topic (panel A) than in response to the worst-matched SEC letter topics (panel C). (Results are not tabulated.) In addition to providing further confirmation of our topic labeling, these results highlight the extent to which SEC questions on a comment result in increased company disclosure on that same topic.

Tables 7 and 8 focus on whether greater scrutiny on certain topics is associated with firm valuation. Table 7 shows eight regressions, one for each SEC letter topic. In each, we regress the price update on the number of words in the SEC letter topic, the number of words in the preliminary prospectus topic, the total number of SEC letters, and control variables previously used in Table 6. The takeaway is that the topics vary in informativeness. After controlling for the aggregate level of SEC review (as proxied by the number of SEC letters), we find that questions related to three topics have incremental explanatory power: revenue recognition, capitalization, and liquidity. A greater focus on each of these topics is associated with significantly lower price updates. Internet Appendix V shows that the total offer size update is similarly strongest for certain topics. Revenue recognition and liquidity are significantly related to both price update and total offer size update.

Table 8 focuses on the extent to which the regulator's information production is unique, that is, the extent to which the SEC uncovers concerns that are not independently discovered by market participants. Recall that while investors

Table 8
What do investors learn from these letters? Looking at CARs around the time of letter publication

A. Relation between topics in the first letter and short-run CARs around the SEC's letter publication							
Dependent variable: CAR (-2, +2) around the SEC's letter publication							
	Topic in the SEC letters	Topic in the SEC letters*	Corresponding topic in S-1	High downloads	# SEC letters	R ²	Obs.
High downloads							
Revenue recognition	0.060 (0.402)	-0.720*** (0.257)	-0.038 (0.113)	3.886** (1.676)	1.154 (0.869)	.063	952
Capitalization	0.240 (0.219)	-0.044 (0.286)	-0.028 (0.116)	0.132 (2.001)	0.988 (0.792)	.060	952
Liquidity	-0.217 (0.299)	-0.270 (0.395)	-0.028 (0.118)	1.327 (2.576)	1.333* (0.777)	.061	952
Risk factors	0.256 (0.378)	0.236 (0.418)	0.029 (0.180)	-1.418 (2.398)	0.915 (0.902)	.061	952
Stock compensation	0.114 (0.665)	-0.266 (0.567)	-0.033 (0.112)	1.308 (2.949)	1.098 (0.789)	.060	952
Executive compensation	0.154 (0.384)	-0.027 (0.604)	-0.025 (0.075)	0.049 (2.898)	1.075 (0.842)	.060	952
Intellectual property	-0.224 (0.606)	-0.046 (0.230)	0.038 (0.183)	0.194 (1.835)	1.193 (0.841)	.060	952
Boiler	-0.253 (0.642)	0.814 (0.526)	-0.027 (0.072)	-3.774 (2.510)	1.104 (0.763)	.061	952
B. Relation between topics in the first letter and long-run CARs after the SEC's letter publication							
Dependent variable: CAR (+3, +90) around the SEC letter publication							
	Topic in the SEC letters	Topic in the SEC letters*	Corresponding topic in S-1	High downloads	# SEC letters	R ²	Obs.
High downloads							
Revenue recognition	-2.203** (0.916)	2.494 (1.915)	0.003 (0.756)	-11.787 (11.674)	-4.039 (4.401)	.102	952

The sample consists of 952 IPOs between 2005 and 2016, for which we download SEC comment letters from EDGAR. In panel A, the dependent variable is cumulative abnormal returns (CARs) measured as the difference between company cumulative returns and cumulative returns on the matched size and book-to-market Fama-French 25 industry portfolio during the (-2, +2) window around the publication of the first letter. High downloads is a dummy variable equals to 1 if the number of letter downloads from the SEC EDGAR Web site during this time interval was higher than the sample median. Information on downloads is taken from SEC log files. Row 1 shows an OLS regression of IPO company CARs on the natural log of the number of words on the revenue recognition topic, this variable interacted with the High Downloads dummy, the number of words in the same topic in the S-1, the High downloads dummy, the number of SEC letters, and controls (company size, number of segments, age, underwriter rank, law firm rank, S-1 length, and S-1 uncertainty, where coefficients on controls are suppressed. Subsequent rows show analogous regressions for each of the other topics. In panel B, the dependent variable is CARs during the (+3, +90) window around publication of the first letter, and we tabulate the regression just for the revenue recognition topic. All regressions include Fama-French 48 industry and year fixed effects, and errors are robust and clustered at the Fama-French 48 industry level.

may infer the number of SEC letters with some precision (using the number of prospectus revisions as a proxy), market participants have no knowledge of the exact content of the letters until at least 20 business days (45 calendar days prior to 2012) after IPOs. For each company, we obtain the date on which the letters are released, and we compute abnormal returns in days (-2, +2) around this date. To account for the effects of investor attention, we also measure the number of times SEC letters were downloaded, in the days around their release. We define high downloads as a dummy equal to one for companies with an above-median number of downloads in the 2 days after the publication of letters.

Results in Table 8, panel A, suggest that SEC concerns regarding one specific topic convey significant negative information to the market. Companies that

receive more SEC concerns related to revenue recognition have significantly lower abnormal returns in the days of the letters' publication. Among companies whose letters were highly downloaded, a 1-standard-deviation increase in SEC attention to revenue recognition results in a significant 0.69% lower abnormal return on the release days. The salience of revenue recognition is consistent with literature across both accounting and finance on the relevance of such issues. Teoh, Welch, and Wong (1998) find that earnings management practices, such as revenue recognition, are related to post-IPO performance; Anderson and Yohn (2002) find these issues are the largest determinant of earnings restatements; Dechow et al. (2011) find that accruals related to revenue recognition are strongly related to fraud; and Kedia and Philippon (2009) link such practices to economic consequences including investment in both capital and labor.

In contrast, we fail to find a significant relation with any of the other topics: coefficients in each of the other regressions are insignificant at conventional levels, the magnitudes are substantially lower, and a bootstrap procedure indicates that the coefficient of interest in each of these other regressions is significantly different than that in the revenue recognition specification. We have also estimated these regressions after excluding cases in which the letters were published within one week of the quiet period expiration, to ensure that this is not contaminating results. Results (not tabulated) are qualitatively similar. In sum, these findings indicate that in addition to not knowing of these revenue recognition concerns at the time of IPOs, investors were unable to completely infer them over the initial weeks of the stock's trading.

Panel B of Table 8 examines in more depth the effects of investor attention, as proxied by the high downloads of the letters. We conjecture that firms with less investor attention at the time of letter release will experience a longer-term drift in abnormal returns, as the revenue recognition concerns contained in these SEC letters gradually disseminate to the market. We regress abnormal returns, measured over the (+3, +90) days following letter release, on (a) our measure of revenue recognition concerns and (b) the interaction of revenue recognition and the high download dummy. Consistent with predictions, the interval over which these revenue recognition concerns are impounded into price is a function of investor attention. Among firms whose letters are highly downloaded, these concerns are incorporated into price faster during the short window around letter release (panel A). Among the less downloaded cases, the concerns are incorporated over the subsequent months (panel B). Finally, similar to short-run CAR results, we find no evidence of a relation between long-run CARs and any topic other than revenue recognition, as reported in Internet Appendix VI.

Table 9 presents similar topic-level analyses, focusing on the extent to which concerns related to revenue recognition are related to other company fundamentals. (Internet Appendix VII provides analogous tests for each of the other topics.) Panel A focuses on measures of firm uncertainty, including initial returns, post-IPO volatility, and post-IPO illiquidity. As established by

Table 9
IPO outcomes and the content of the first SEC comment letter

A						
	<i>Initial return</i>	<i>Volatility, first year</i>	<i>Amihud's illiquidity, first month</i>	<i>Amihud's illiquidity, first year</i>		
Revenue recognition	−1.590 (1.759)	0.031*** (0.008)	0.002 (0.002)		0.007** (0.003)	
Controls	Yes	Yes	Yes		Yes	
Year FE	Yes	Yes	Yes		Yes	
Industry FE	Yes	Yes	Yes		Yes	
R ²	.175	.458	.398		.513	
Observations	952	952	952		952	
B						
	<i>Secondary shares (dummy)</i>	<i>Secondary shares offers</i>	<i>Overallotment exercised</i>	<i>Insider sales</i>	<i>Abnormal return, first half-year</i>	<i>Abnormal return, first year</i>
Revenue recognition	0.039** (0.020)	3.834** (1.745)	−0.765** (0.315)	0.044 (0.031)	−3.149** (1.279)	−6.625** (3.071)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	.317	.305	.221	.138	.082	.098
Observations	952	952	952	952	952	952

The sample consists of 952 IPOs between 2005 and 2016, for which we download SEC comment letters from EDGAR. In panel A, we regress measures of post-IPO uncertainty on the natural log of the number of words in the revenue recognition topic, within the first SEC letter. The dependent variables are initial return (Column 1), volatility measured over the first year after the IPO (Column 2), Amihud's illiquidity measured over the first month after the IPO (Column 3), and Amihud's illiquidity measured over the first year after the IPO (Column 4). Panel B is similar, but the dependent variables are a dummy equal to one if any secondary shares were offered in the IPO, zero otherwise (Column 1), the percentage of total shares offered that represented secondary shares (Column 2), overallotment shares as a percent of total shares (Column 3), Insider sales (Column 4), Abnormal returns over the first 6 months after the IPO (Column 5), and Abnormal Returns over the first year after the IPO (Column 6). All regressions include Fama-French 48 industry and year fixed effects, and errors are robust and clustered at the Fama-French 48 industry level.

Rock (1986) and Beatty and Ritter (1986), offerings with greater uncertainty and higher information asymmetry will tend to have higher initial returns. Volatility of stock returns captures uncertainty in underlying firm valuation over the year following the offer. Finally, as discussed by Amihud (2002), stocks with greater information asymmetry are characterized by lower liquidity, which we capture with the Amihud illiquidity measure, over both the 1-month and 1 year following IPOs.

Panel B focuses on measures of firm valuation. First, insider selling at the time of IPOs can represent a negative signal regarding future firm performance, and, thus, we include two measures of secondary sales at the time of IPOs: a dummy equal to one if there were any insider sales, and secondary shares as a percentage of total shares sold. Second, external investors' demand for IPO shares represents a signal regarding their valuation of the company. Thus, we include the size of the overallotment option that was exercised, which represents a measure of investor demand. Third, we examine insider sales around the time of the lockup expiration, to capture insider perceptions regarding the company at this point in time. Finally, we include abnormal returns over both the first 6 months following IPOs and the 1 year following IPOs.

Table 10
Changes in disclosure for withdrawn and completed IPOs

A. Multivariate analysis

	Dep't var = # SEC Letters before price range	Dependent variable: Withdrawn IPO dummy	
# S-1 amendments before the price range	0.511*** (0.033)		
Change in disclosure		-0.954*** (0.295)	
# letters		0.039*** (0.006)	0.022*** (0.008)
Revenue recognition			0.037*** (0.014)
Revenue recognition in S-1			0.138*** (0.010)
Nasdaq returns	-0.004 (0.006)	-0.005*** (0.002)	0.004 (0.003)
Sample	All completed IPOs + withdrawn IPOs with SEC letters	All completed and withdrawn IPOs	Withdrawn IPOs + matched sample of completed IPOs
Observations	1,016	1,071	200
Industry, year FEs	Yes	Yes	No
Controls	Yes	Yes	Yes
R ²	.521	.341	.876

B. Univariate analysis

	Change in disclosure, % word increase from S-1 to first S-1/A containing initial price range		
	Withdrawn IPOs	Completed IPOs	t-statistics of difference
Full sample	8.20 (n = 64)	11.16 (n = 952)	-3.00***
One letter	7.31 (n = 11)	7.18 % (n = 102)	0.07
Two letters	6.13 (n = 21)	10.03 (n = 286)	-2.99**
Three letters	8.37 (n = 14)	10.80 (n = 282)	-0.88
Four or more letters	11.01 (n = 18)	14.22 (n = 276)	-1.62

The sample consists of the 952 completed IPOs used in prior analyses plus 119 withdrawn IPOs, 64 of which we have actual SEC letters and 55 of which we do not. In panel A, Column 1 shows a predictive model of the number of letters across the 952 completed IPOs and the 64 withdrawn IPOs for which we have the actual letters. Coefficients from this model combined with company characteristics of the 55 withdrawn IPOs for which we do not have actual letters yield the estimated number of letters for this sample, $\widehat{Letters}$. Column 2 shows a regression of withdrawal (defined as a dummy equal to one if the offering was withdrawn, zero otherwise) on the number of SEC letters and the change in disclosure. For the subset of withdrawn offerings where we do not have the actual number of letters, we use $\widehat{Letters}$ (from the regression in Column 1). Change in disclosure is defined as $\ln(\text{number of words in first amended prospectus with a price range} / \text{number of words in preliminary prospectus})$. In Column 3, we regress the withdrawal dummy on the natural log of the number of words in the first SEC letter pertaining to revenue recognition. Column 3 is based on a matched sample, where for each withdrawn IPO we select up to three completed IPOs in the same year and Fama-French 48 industry with the closest propensity score; the propensity score is based on offer size, number of segments, length of S-1, and Nasdaq returns. All columns include Nasdaq returns over the 2 months following the filing date, as well as company controls (not tabulated). The first two columns also control for year and Fama-French industry fixed effects. All errors are robust and clustered at the Fama-French 48 industry level. In panel B we limit the sample to IPOs with the letters available (952 completed, 64 withdrawn), and we report univariate tests comparing the change in disclosure (expressed as a percentage) across completed and withdrawn IPOs.

We regress these measures of both uncertainty and valuation on the log of the number of words on revenue recognition, plus control variables used in prior analyses. Control variables include the number of words on the associated topic within the prospectus, the number of SEC letters, and company and offer-specific factors used in prior regressions. We only report the coefficient on the variable of interest for each regression.

The key takeaway from Table 9 is that SEC questions related to revenue recognition are significantly related to many of these metrics of company uncertainty and valuation. Looking first at panel A, companies with greater concerns related to revenue recognition have significantly higher post-IPO volatility and illiquidity. The extent of the revenue recognition questions in the SEC letters is economically significant: 1-standard-deviation increase in these concerns is associated with 11.7% standard deviation higher volatility and 12.6% standard deviation greater illiquidity during the first year.

Panel B highlights the extent to which these concerns also represent negative signals regarding company valuation. In addition to having lower price updates, companies that receive more questions related to revenue recognition have lower post-IPO abnormal returns over both the 6-month and 1-year interval, consistent with the company being worth less than was originally expected. Insiders appear to be aware of these effects. At the time of their IPO, companies with greater concerns related to revenue recognition are 4.0% more likely to have secondary shares sold, and tend to have a greater percent of secondary shares sold. In addition, these offerings experience lower levels of demand from external investors, as captured by the negative relation with the portion of the overallotment option that is exercised.

In comparison, as shown in Internet Appendix VII, we find no evidence that any of the other topics are significantly related to any of these metrics of company uncertainty and valuation. This is consistent with the abnormal returns around the public release of the SEC letters (shown in Table 8), indicating that revenue recognition is uniquely informative regarding the future prospects of the company.

We also examine the relation between the incidence of post-IPO class action lawsuits and SEC concerns on each of these topics. Across our sample of 952 IPOs, 51 were sued in Section 11 class action lawsuits within 1 year of the offer date (see, e.g., Lowry and Shu 2002). Comparing the SEC letters of the two groups of companies, we find that companies that were ultimately sued had significantly greater revenue recognition-related concerns. In contrast, we find no significant difference in attention to any of the other topics. Moreover, in a series of both univariate and multivariate regressions, SEC attention to revenue recognition is one of the strongest predictors of post-IPO lawsuits. (Internet Appendix VIII shows the results.)¹⁸

¹⁸ SEC letters could possibly contribute to lawsuits; for example, they could be used as court evidence of these firms' questionable conduct.

In aggregate, results in Tables 7–9 suggest that revenue recognition concerns are unique in the strength of their association with post-IPO changes in firm value and with post-IPO uncertainty and liquidity. Moreover, they appear to not be independently fully recognized by investors. The strong relation between revenue recognition concerns and firm value is consistent with both the SEC's focus on this issue and with prior literature.¹⁹ DeChow, Lawrence, and Ryans (2016) find that concerns related to revenue recognition are associated with both more insider selling around the public release of SEC letters pertaining to mature firms' 10K filings and a negative return drift over the following 2 to 3 months. They conclude that the market's failure to more quickly impound this information into prices reflects investor inattention, as investors download the SEC letters relatively rarely. Our findings suggest that even among a set of companies in which inattention might be less of an issue, investors are slow to recognize these revenue recognition-related issues. Importantly, our findings are consistent with the regulator playing a unique role, producing information that investors do not.

2.3 Withdrawn offerings

The SEC information discovery process uncovers firm-specific information (and requires the firm to publicly disclose such information) at a critical point in time – precisely when the firm is deciding whether to plough ahead with the going public process. This section focuses on the possibility that these demands for information revelation influence a company's decision to withdraw its IPO. Negative information revealed through the process may cause the company and its underwriter to determine that its valuation would be lower than previously estimated. Alternatively, the company may simply determine that the costs of revealing all the necessary information are too high, for example, for competitive reasons.

Although the SEC process might reveal valuable information about the likelihood and the reasons for withdrawals, the information we have for this subsample is more limited than what we are able to obtain for the general sample. Unlike for completed IPOs, where the SEC publicly releases the letters via EDGAR, for the universe of withdrawn IPOs, they do not do so. Thus, neither the public nor researchers have been able to access the content, or even the number, of letters written to companies whose IPOs were eventually withdrawn. We were able to obtain letters for 64 of the 119 withdrawn IPOs during our sample period through a FOIA request.²⁰

¹⁹ See, for example, overviews by both the SEC (<https://www.sec.gov/divisions/corpfin/fortune500rep.htm>) and practitioners. PricewaterhouseCoopers features crucial revenue recognition concerns in their reports regarding all industries they review, including technology, media, health, and consumer products. (<https://www.pwc.com/us/en/cfodirect/publications/sec-comment-letter-trends.html>).

²⁰ Time constraints at the SEC contributed to limitations in our analysis, because we were only able to receive the letters for a subset of the withdrawn offerings. We have no reason to believe that the sample is biased, a conjecture

Our main prediction is that the likelihood of withdrawal is higher among firms that receive heavier demands for additional disclosure and among firms who revise their disclosure less in response to SEC requests. We begin by measuring demands for disclosure as the total number of SEC letters, and disclosure revisions as $\ln(\text{the number of words first amended prospectus with initial price range} / \text{the number of words in the initial prospectus})$.

Recognizing that we only have our proxy for SEC demands for disclosure (i.e., number of letters) for a subset of the withdrawn IPOs, we do the following. We use the actual number of letters for the 952 completed IPOs + 64 withdrawn IPOs where we have this information. For the remaining fifty-five withdrawn IPOs we include the estimated number of letters. This estimate is based on a predictive model, shown in Column 1 of Table 10, panel A, where we regress the number of letters (of each company for which we have actual letters) on the number of prospectus revisions, Nasdaq returns over the 2 months after the filing date, and firm characteristics. Coefficients from this model combined with actual firm and offer characteristics yield the estimated number of letters, *Letters*, for the fifty-five withdrawn offerings where we lack actual letters. Essentially, we estimate the number of letters as a function of the number of prospectus revisions, conditioning on firm, IPO, and market characteristics.

Column 2 of Table 10 shows our first main result. We include SEC disclosure requests (# letters when this information is available, and *Letters* in the other fifty-five cases) and the company's change in disclosure as the independent variables of focus, and the dependent variable equals a dummy equal to one if the company withdrew its IPO. Consistent with predictions, we find that companies with a greater number of SEC letters, and companies that increase disclosure more in association with SEC requests have a lower likelihood of withdrawal. One more SEC letter is associated with a 3.9% lower completion rate, and a 1-standard-deviation greater increase in disclosure is associated with a 4.2% higher completion rate. It is noteworthy that the economic magnitude of change in prospectus revisions in explaining IPO withdrawal is more than that of Nasdaq returns, which have been highlighted in prior literature as one of the most important determinants of withdrawal (e.g., Dunbar 1998; Bernstein 2015).²¹

Column 3 examines the extent to which the relation between SEC concerns and IPO withdrawal is related to SEC concerns about revenue recognition. Because we need the actual letters to analyze topic content we can only use

that (untabulated) descriptive statistics support. The sample of withdrawn IPOs is subject to the same criteria as completed IPOs; for example, it excludes REITs and unit offerings. We also required the lower end of the first price range to be above \$5.00 for the withdrawn IPO sample.

²¹ In our sample (2005–2016), a 1-standard-deviation increase in Nasdaq returns lowers withdrawal chances by 2.5%, compared to Bernstein (2015), who reports a 8.72% decrease in the likelihood of a firm completing its IPO. The difference could be at least partially explained by the differences in the standard deviation of Nasdaq returns between the two sample periods: 5% during our sample period and 12% during Bernstein's sample period (on average in the months leading up to each completed IPO).

the sixty-four withdrawn IPOs with the actual SEC letters available. Given the very small sample we are unable to include industry and year fixed effects and therefore we use a matched sample approach, restricting the control firms to be in the same industry and year as firm in the treated sample. We then estimate a propensity score based on the four most significant determinants of withdrawal: offer size, number of segments, S-1 length, and Nasdaq returns.²² For each withdrawn offering, we then select up to three completed IPOs with the closest propensity score. Finally, in Column 3, we regress the IPO withdrawal dummy on the natural log of number of words related to revenue recognition plus all controls. Consistent with results throughout the paper, more concerns related to revenue recognition represents a negative signal: these companies are significantly more likely to withdraw the offering. In contrast, we find no similar relation for the other topics (shown in Internet Appendix X).

Panel B presents an alternative, more simplistic approach toward analyzing the number of letters. In this approach, we focus on IPOs with available letters (952 completed, 64 withdrawn). Consistent with conclusions from panel A, we see that relative to completed IPOs, withdrawn IPOs revise their prospectus less. Across the full sample, completed IPOs have an 11.16% increase in disclosure, compared to 8.20% among the withdrawn sample, significant at the 1% level. Moreover, we observe this same pattern within almost every subsample of IPOs, that is, IPOs with two letters, three letters, and so forth.

Our findings are consistent with the notion that management of companies that ultimately withdraw their IPOs feel less comfortable fully responding to SEC concerns, compared to management of firms that complete their IPOs. This potentially reflects lower management quality, a greater perceived amount of proprietary information, or more negative material that management had been seeking to keep private.²³ This finding highlights the relation between the regulator's information production and firms' decisions to withdraw their IPOs. The regulator cannot bar a company from going public. However, the regulatory demands for information disclosure can result in companies opting to stay private.

3. Discussion

Our findings throughout the paper indicate that the SEC review process highlights issues that are relevant to underlying company value and that investors do not independently uncover. Regulator concerns around this key capital raising event are related to revisions in the offer price, the

²² As shown in Internet Appendix IX, the withdrawn sample and matched sample are similar along these four dimensions. The only dimension significantly different between the two samples is the number of segments, and the difference is small in economic terms (1.0 vs. 1.1).

²³ Although these results also would be consistent with the SEC simply asking withdrawn IPOs fewer questions in their letters, we can think of no reason this would be the case, and we find no evidence to support this conjecture: completed IPOs receive an average 41 question in their first letter, compared to 44 questions for withdrawn IPOs.

likelihood of IPO withdrawal, insider sales both in IPOs and around the lockup expiration, post-IPO liquidity, and post-IPO abnormal returns. All of these dynamics suggest that firms that are scrutinized more by the SEC, particularly on the topic of revenue recognition, are more opaque and have weaker prospects. Transaction costs also increase, as the SEC scrutiny prolongs the time firms spend in the process. In a Coasian world, there would be no role for the regulator (Coase 1960). In contrast, the extent of uncertainty and high costs of acquiring information on private companies arguably make the regulator's role particularly relevant. The regulator can both demand information and impose enforcement criteria, in ways that other market participants cannot (see La Porta et al. (2006) for a discussion). Given the public-good nature of information, the regulator's actions have value: they increase transparency for potential investors and arguably improve resource allocation.

Importantly, the benefits of this increased transparency are lessened by the fact that the regulator's information is not made public until several weeks after IPOs. Interestingly, the SEC comment letter policies have moved in the direction of increased transparency. The SEC first began publicly releasing the letters on EDGAR in 2005, with the release occurring a minimum of 45 days after IPOs. In 2012, it began disclosing the letters more quickly after IPOs, specifically beginning 20 days instead of 45 days after the offering. In each case, the stated reason for the change was increased transparency. Nonetheless, our findings raise the question of why investors do not have access to this information even sooner, particularly given the fact that one of the SEC's stated objectives is to "protect investors."

Informal conversations with the SEC suggest that the Commission does not release the letters at the time of their writing (i.e., prior to IPOs) as a way to protect companies. Reasons include not influencing the marketing of the offer, not having undue impact on company valuation, and not contributing to higher stock volatility on a day that is already characterized by extremely high trading and price movement. In addition, the SEC may seek to protect companies that ultimately withdraw their offerings.²⁴ Companies that remain private (perhaps) should have the right to keep various information private, that is, to not publicly disclose all information contained in the letters. Interactions with the SEC concerning the withdrawn IPOs' data are consistent with this notion. In our first few attempts to obtain the letters of withdrawn IPOs through a FOIA request, a sticking point was the fact that these companies are private and thus the information should not be publicly released (though the SEC ultimately determined that this concern was insufficient to prevent sharing of the information under the FOIA and did release these letters, as discussed in the prior section).

²⁴ In conversations with the SEC, it was emphasized that the letters would be released a minimum of 20 days after the offer, with the SEC retaining the right to delay release longer. It was also noted that this policy was based on staff guidance, and it was not an official rule made by a Commissioner.

Whether companies' "right to privacy" outweighs the benefits of increased transparency, less asymmetric information and fair pricing is an issue the SEC might want to reconsider. One possible approach might be to publicly release the SEC letters, but with particularly sensitive portions redacted. For example, if a company kept information regarding a pending patent out of the public domain for reasons of competitive advantage (sharing the information only with the SEC and with relevant references to the patent redacted within the prospectus), then any questions in the SEC letters relating to this patent would be similarly redacted.

Finally, given recent advances in technologies (like the ones used in this paper), more sophisticated investors can extract this valuable information, at least in part, by analyzing the changes in the prospectus revision during the IPO process. Our results suggest that detecting significant changes related to revenue recognition has investment value, both around the time of the letters' release as well as for predicting longer term performance. As a by-product, the use of such strategies by some market participants will also result in the SEC's policy of delaying the release of the letters becoming less relevant as the information, to an extent, may already be impounded into price.

4. Conclusion

The IPO process is complex, with many players taking an active role. There is a large literature on the role of the issuer, underwriter, analysts, market makers, auditor, and lawyers. The regulator also plays a substantial role in the IPO process, but there is limited information on how the SEC fulfils its regulatory role, and how it influences companies as they prepare to go public. This paper strives to fill some of this gap.

We find that the SEC is actively involved in each IPO, communicating with the company an average 4 times between the filing of the preliminary prospectus and the IPO date. The intensity of SEC review has an effect on the amount of material a company discloses. On the other hand, these dynamics prolong the pre-IPO process, effectively increasing the transaction costs associated with going public. The longer time spent in registration can make the IPO process more costly and risky for the company. Further, our analysis suggests that firms who are more reluctant to provide the SEC with requested information are more likely to withdraw their IPOs.

To gain more insight into the details of the SEC process, we use LDA techniques to extract the main subjects of the SEC review and to quantify the extent of each topic in each letter. We develop a technique (combined use of the LDA and KL divergence) that enables us to analyze and contrast information produced by both the company through its prospectuses, and by the regulator in a series of letters. Importantly, we also develop a method to objectively assign an economic meaning to each topic. These techniques should have applications

in a variety of other contexts, within finance as well as accounting and other fields.

Our analysis reveals that the SEC focuses its attention on topics that largely correspond to its regulatory mandate. Across all topics, questions related to revenue recognition are the most informative about future company prospects. Companies that receive more questions on this topic have significantly lower price updates prior to their IPO, higher insider sales at both their IPO and at the expiration of the lockup period, higher volatility and illiquidity after their IPO, and lower post-IPO performance. Consistent with revenue recognition issues being particularly important, the abnormal return at the time the SEC letters are publicly released is significantly negative for companies that receive letters with more attention to this topic.

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Appendix. Variable Definitions

A.1 Textual Analysis Variables

Throughout the paper, we define “English words” as words that are contained in Grady Ward’s English Wordlist; the full list of 122,806 words is available as a part of the “qdapDictionaries” package in R. In the analysis we use only word-roots that are extracted with the “SnowballC” R-package. As is common in most textual analyses, common words (typically referred to as stopwords), for example, as “a” and “the,” are excluded.

We use an eight-topic LDA analysis of the first SEC letter relating to an IPO’s prospectus that each company receives. Topics produced by this analysis are labeled as Revenue Recognition, Capitalization, Liquidity, Risk Factors, Stock Compensation, Executive Compensation, Intellectual Property, and Boilerplate. For each of these eight topics we measure three variables which capture: the extent of the topic in the first SEC letter, extent of the most similar topic in the first version of the prospectus, and change in the most similar topic from the first prospectus to the last.

Here, we provide more details regarding how each variable is measured for Revenue Recognition. The method is the same for other topics.

A.1.1 Number of revenue recognition words in the first SEC letter. This variable is defined as: $\ln(\% \text{ of topic in the letter} \cdot \text{number of words in the letter})$, where *% of topic in the letter* is the direct output of the LDA analysis (showing % of the letter that is devoted to the topic of interest),

Table A1
Company variables

Variable name	Variable description	Data source
<i>Age</i>	Age is measured as a natural logarithm of (1 + Firm age), where Firm age is the difference between the firm's year of IPO and its foundation year	Jay Ritter's Web site, IPO prospectuses
<i># of segments Underwriter rank</i>	Number of segments during the previous fiscal year Underwriter rank, measured as in Loughran and Ritter (2004)	Compustat Jay Ritter's Web site
<i>Law firm rank</i>	Defined as the number of IPOs a given law firm worked on in the 5 years before the IPO of interest	SDC Platinum
<i>S1 length</i>	Natural logarithm of the total number of English words in the main file of the first version of an IPO's prospectus, the S1. English words are defined as words from Grady Ward's English Wordlist; the full list of 122,806 words is available in the "qdapDictionaries" package in R	SEC EDGAR
<i>S1 uncertainty</i>	(Number of uncertain words/total number of English words) in S1, where the number of uncertain words is the sum of all uncertain, negative, and weak words in the first prospectus (see Loughran and McDonald 2013)	SEC EDGAR, Bill McDonald's Web site
<i>High F-score</i>	Dummy variable equal to 1 if the F-score is above 1 (higher than average risk of accounting misstatements). The F-score, a likelihood of accounting misstatement measure, is defined as in Dechow et al. (2011)	Compustat
<i>Company size</i>	Natural logarithm of Compustat SALE variable one fiscal year prior to an IPO	Compustat
<i>VC</i>	Dummy equal to 1 if SDC VC variable = Yes	SDC Platinum
<i>JOBS Act</i>	Dummy equal to 1 if the company files its IPO under the JOBS Act, as stated in the prospectus	SEC EDGAR
<i>Number of words in a topic, in first SEC letter</i>	Natural logarithm of the number of English words in the topic, in Part 4 (Itemized part) of first SEC IPO-related comment letter. The number of words is measured as the length of the letter * % of letter dedicated to each topic, where the percentage comes from the LDA output	SEC EDGAR
<i>Corresponding topic in first S1</i>	Natural logarithm of the number of English words in the topic, in the main body of the first prospectus, the S1. The number of words is measured as the length of the S1 * the percentage of S1 dedicated to each topic, where the percentage comes from the LDA output. The matching of topics between the SEC letters and the S1 is determined by the KL divergence	
<i># SEC letters</i>	Natural logarithm of the number of SEC IPO-related comment letters (SEC form UPLOAD). Whether or not a given comment letter is IPO-related is determined based on the SEC film number in the title of the letter. The SEC film number for IPO-related letters refers to the first prospectus. Letters that do not include this number are manually checked. In the analysis of withdrawn offerings, we focus on the period prior to the setting of the price range. We use number of letters as a variable if we were able to obtain them through FOIA require or <i>Letters</i> – the estimate the number of letters based on a predictive model (Column 1 of Table 10) for the subset of withdrawn offerings where we do not have actual letters	SEC EDGAR
<i># SEC letters before the price range</i>	Natural logarithm of (1 + number of SEC IPO-related comment letters after before price range is set). Information about the timing of the price range is obtained from SDC Platinum price range history variable. Here, we use the natural logarithm of one plus the number of letters, to account for the small number of cases in which the number of letters prior to the setting of the price range equals zero	SEC EDGAR, SDC Platinum

Table A1
continued

Variable name	Variable description	Data source
<i>Registration period length</i>	Natural logarithm of the number of days between an IPO issue date and the filing date	SDC Platinum
<i>Change in disclosure</i>	Natural logarithm of (#words in 424/ #words in S1). In topic-level analyses, this is defined analogously at the topic level. In the analysis of withdrawn offerings, we focus on the increase from the first prospectus to the first prospectus with the price range, defining the variables as $\ln(\text{number of words in first amended prospectus with a price range} / \text{number of words in preliminary prospectus})$	SDC Platinum
<i>Price update</i>	Percentage change from the midpoint of the first price range to the office price	SDC Platinum
<i>Shares update</i>	Percentage change from the first number of shares offered to the final number of shares offered	SDC Platinum
<i>Size update</i>	Percent chance from initial size (first number of shares * midpoint of the first price range) to the final number of shares offered * offer price	SDC Platinum
<i>Initial returns</i>	Percentage change from the offering price to the closing price	CRSP, SDC Platinum
<i>Volatility</i>	Annualized standard deviation of the daily stock returns in the interval return between 21 and 273 trading days after the IPO date	CRSP
<i>Amihud's illiquidity, first month</i>	Average value of Amihud's illiquidity in the first 30 days following the offering. Amihud's illiquidity calculated as $10^6 * \text{abs}(\text{daily returns}) / (\text{daily volume} * \text{daily price})$	CRSP
<i>Amihud's illiquidity, first year</i>	Average value of Amihud's illiquidity in the first 365 days following the offering	CRSP
<i>Secondary shares (dummy)</i>	Dummy variable which equals to 1 if any secondary shares were offered	SDC Platinum
<i>Secondary shares offers</i>	Percentage of shares offered as secondary shares at IPO	SDC Platinum
<i>Overallotment exercised</i>	Percentage of shares exercised as overallotment options	SDC Platinum
<i>Insider sales</i>	Dummy variable which equals to 1 if there were any insider sales in the company in the window between 170 and 190 calendar days after the offer date	Thompson Reuters
<i>Abnormal return, first half-year</i>	Difference between company stock return and corresponding size and book-to-market 25 industry portfolio returns during the 175 calendar days after an IPO	CRSP, Kenneth French's Web site
<i>Abnormal return, first year</i>	Difference between company stock return and corresponding size and book-to-market 25 industry portfolio returns during the 365 calendar days after an IPO	CRSP, Kenneth French's Web site
<i>CAR (-2, +2)</i>	Abnormal return during (-2, +2) days around the publication of the first SEC comment letter on the SEC EDGAR platform, where abnormal return is defined as the company's return minus the return on the corresponding size and book-to-market Fama-French 25 industry portfolio	CRSP, Kenneth French's Web site, SEC log files
<i>CAR (+3, +90)</i>	Abnormal return during (+3, +90) days around the publication of the first SEC comment letter on the SEC EDGAR platform, where abnormal return is defined as the company's return minus the return on the corresponding size and book-to-market Fama-French 25 industry portfolio	CRSP, Kenneth French's Web site, SEC log files
<i>High downloads</i>	Dummy variable equal to one if the first SEC letter has more downloads in the first 2 days after its publication than a median SEC IPO-related comment letters in the first 2 days after its publication. We count the number of downloads as the number of unique noncrawler IP addresses that downloaded the letters in this period	SEC log files
<i>Nasdaq return</i>	Return of Nasdaq index measured over the 2 months after an IPO's (completed or withdrawn) filing date	CRSP

and. *number of words in the letter* is the number of English words (according to Grady Ward's Wordlist) in the itemized part (Part 4 in Internet Appendix II) of the first SEC letter.

LDA analysis is performed with LDA() function of the topicmodels package in R. In preparation for the LDA analysis we:

1. Exclude words that are not in Grady Ward's Wordlist (this list is available in the "qdapDictionaries" package in R)
 - This step excludes non-English words and names from the text.
2. Exclude stopwords, defined as words that belong to the stopwords ("english") list in the tm package in R.
 - This step excludes very common and uninformative words (e.g., "end," "or," and days of the week)
3. Extract root words from all words with the stemDocument function in the SnowballC package in R.
 - This conversion allows us to group words with the same roots together.
4. Exclude rare words.
 - To simplify calculations, we limit our analysis to the set of root words that appear in at least five different first letters.
5. The final list of root words for the LDA analysis includes 4,420 different root words. All LDA and HDP analyses are based on this list.

A.1.2 Revenue recognition topic in the first S1. We use a 28-topic LDA analysis on the first prospectus of each company, the S1. To select the topic in the prospectus that corresponds to the Revenue Recognition topic in the first SEC letter, we use KL divergence.

We calculate the KL divergence between the Revenue Recognition topic in the SEC letter and each of the 28 S1 topics, a total of 28 KL divergences. The KL divergence represents the amount of incremental information that the Revenue Recognition topic in the SEC letter adds, relative to the matched S1 topic.

More specifically, if $Q(w)$ is the saliency of word w in the Revenue Recognition topic and $P_j(w)$ is the saliency of word w in the S1 topic j ($j \in \{1, \dots, 28\}$) then the KL divergence between these two topics would be calculated according to the following formula:

$$KL_j = \sum_{w \in \text{all words}} Q(w) \cdot \log \frac{Q(w)}{P_j(w)}.$$

The prospectus topic that minimizes the value above is defined as the Revenue Recognition topic in the prospectus.

Lastly, the Revenue Recognition Topic in the Prospectus is defined as

$\ln(\% \text{ topic in S1} \cdot \text{number of words in S1})$,

where $\% \text{ of topic in S1}$ is the direct output of the S1 LDA analysis (showing the % of the document that is devoted to the topic of interest) and $\text{number of words in S1}$ is the number of English words (according to Grady Ward's Wordlist) in the first prospectus.

A.1.3 Change in disclosure on the revenue recognition topic. To measure the change in disclosure in each topic, we utilize the 28-topic LDA analysis from the first prospectuses and apply it to the set of final prospectuses.

Hence, we have two LDA outputs: 28 LDA topics on the S1s and 28 LDA topics on the 424s. Because we have applied an already existing LDA to the 424s, the topics in the S1s and 424s are the same.

Table A2
Titles of SEC letters paragraphs

Paragraph name	Number of paragraphs extracted	Percentage of SEC letters with this paragraph
Risk factors	769	21.36%
Business	703	19.53%
Prospectus summary	635	17.64%
Use of Proceeds	568	15.78%
Results of operations	412	11.44%
Liquidity and capital resources	392	10.89%
Management	386	10.72%
Overview	343	9.53%
Revenue recognition	281	7.81%
Executive compensation	277	7.69%
Underwriting	254	7.06%
Certain relationships and related party transactions	244	6.78%
Principal and selling stockholders	205	5.69%
Description of capital stock	194	5.39%
Capitalization	192	5.33%
Dilution	176	4.89%
Compensation discussion and analysis	175	4.86%
Stock-based compensation	175	4.86%
Critical accounting policies and estimates	169	4.69%
Intellectual property	165	4.58%
Principal stockholders	164	4.56%
Competition	157	4.36%
The offering	154	4.28%
Our company	105	2.92%
Contractual obligations	101	2.81%

This table presents the twenty-five most common paragraph titles in the SEC letters. The second column represents the total number of paragraphs under this title, as extracted from the letters. The third column shows the percentage of SEC comment letters that have this paragraph.

We measure change in disclosure according to the following formula:

$$\ln \left(\frac{\% \text{ topic in 424} \cdot \text{number of word in 424}}{\% \text{ topic in S1} \cdot \text{number of word in S1}} \right).$$